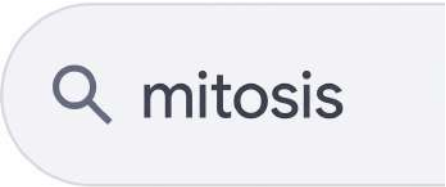
Feb 2026

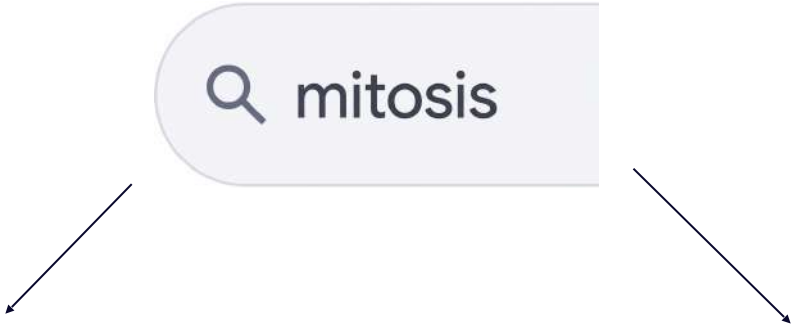
Improving Search Ranking with LambdaMART and Elasticsearch

Brandon (Anbang) Wu

BM25



BM25



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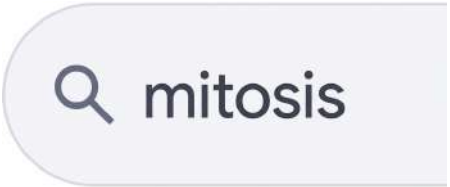
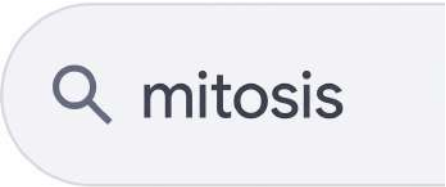


(1) Term Frequency

How often “mitosis” appears in a document

BM25



(1) Term Frequency

How often “mitosis” appears in a document

(2) Inverse Document Frequency

How rare “mitosis” is across all documents

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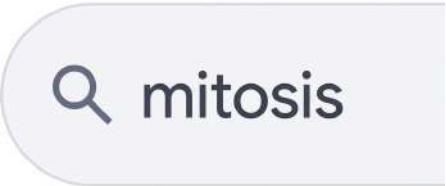

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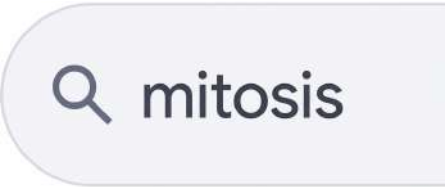

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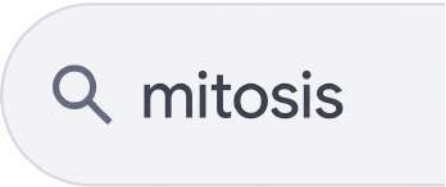
BM25 Score
per document

BM25



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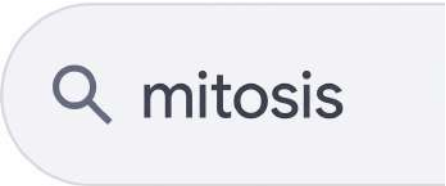

(1) Term Frequency

How often “mitosis” appears in a document

(2) Inverse Document Frequency

How rare “mitosis” is across all documents

BM25 Score
per document

Pick Highest ones

BM25
Is Not Enough

User-generated content
900M+ flashcard sets

Homogeneous content
with identical titles
from different users

BM25 scores very close

mitosis

Generate

Study sets

9 studiers now
Mitosis and Meiosis
25 terms • 4.5 (417)
Preview

92 studiers today
Mitosis
9 terms • 4.8 (87)
Preview

157 studiers today
Mitosis
26 terms • 4.5 (52)
Preview

41 studiers today
Mitosis
13 terms • 4.7 (41)
Preview

Set preview

Mitosis and Meiosis

Study

interphase
phase before mitosis; growth of cell and preparation for division

prophase
chromatin condenses into chromosomes, nuclear envelope disappears, spindle fibers start to form

G1
growth and functioning

Synthesis
DNA replicates

G2
organelles replicate in preparation for division

metaphase
chromosomes, chromosomes line up in the middle of the cell

anaphase
the sister chromatids seperate into individual chromosomes and are moved apart

telophase
chromosomes return to chromatin; nuclear membranes form; cleavage furrow begins

BM25
Is Not Enough

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900M+ flashcard sets

Homogeneous content
with identical titles
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BM25 scores very close

mitosis

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Extra Signals For Elastic Boosting

In addition to BM25 scoring on Titles and Text Blobs

Boosted Term Query

```
{
  "term": {
     "Field A": {
       "value": 1,
       "boost": 3
    }
  }
}
```

If the document's field A value is 1,

multiply its score by 3.

Extra Signals For Elastic Boosting

In addition to BM25 scoring on Titles and Text Blobs

Function Score Query

```
{
  "field_value_factor": {
    "field": "Field B",
    "factor": 1,
    "modifier": "ln1p"
  }
}
```

→

Popularity boost

Boost the document score by the natural log of Field B

Extra Signal Boosting

In addition to BM25 scoring on Titles and Text Blobs

After years of manual tuning driven by heuristics:

Manual tuning showed diminishing returns.

We can no longer rely solely on heuristics to achieve meaningful breakthroughs.

Proposed Solution: Learning-to-Rank

https://www.elastic.co/docs/solutions/search/ranking/learning-to-rank-ltr

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https://www.elastic.co/docs/solutions/search/ranking/learning-to-rank-ltr

Search context

text_query: foo

Proposed Solution: Learning-to-Rank

https://www.elastic.co/docs/solutions/search/ranking/learning-to-rank-ltr

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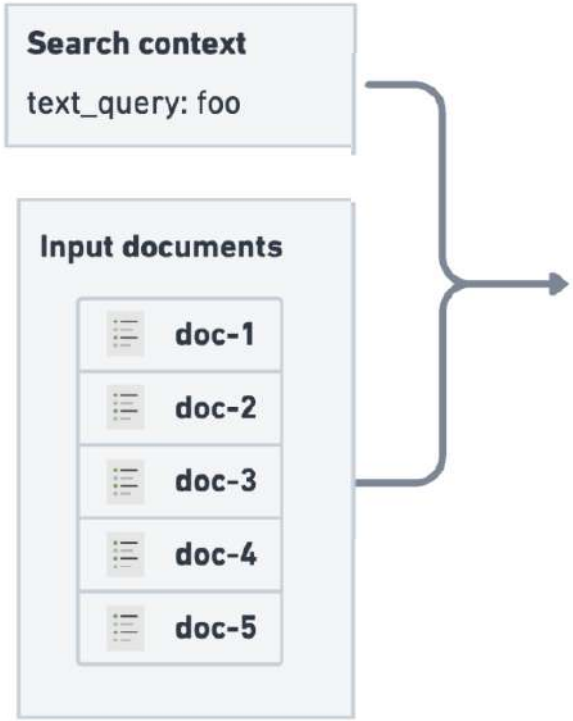
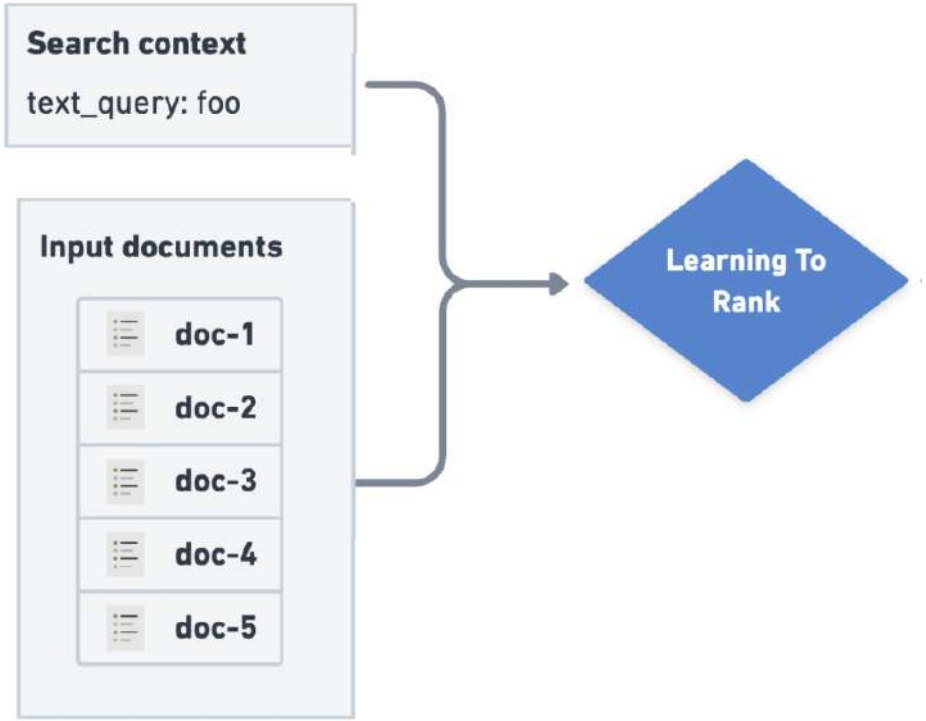


Proposed Solution: Learning-to-Rank

https://www.elastic.co/docs/solutions/search/ranking/learning-to-rank-ltr



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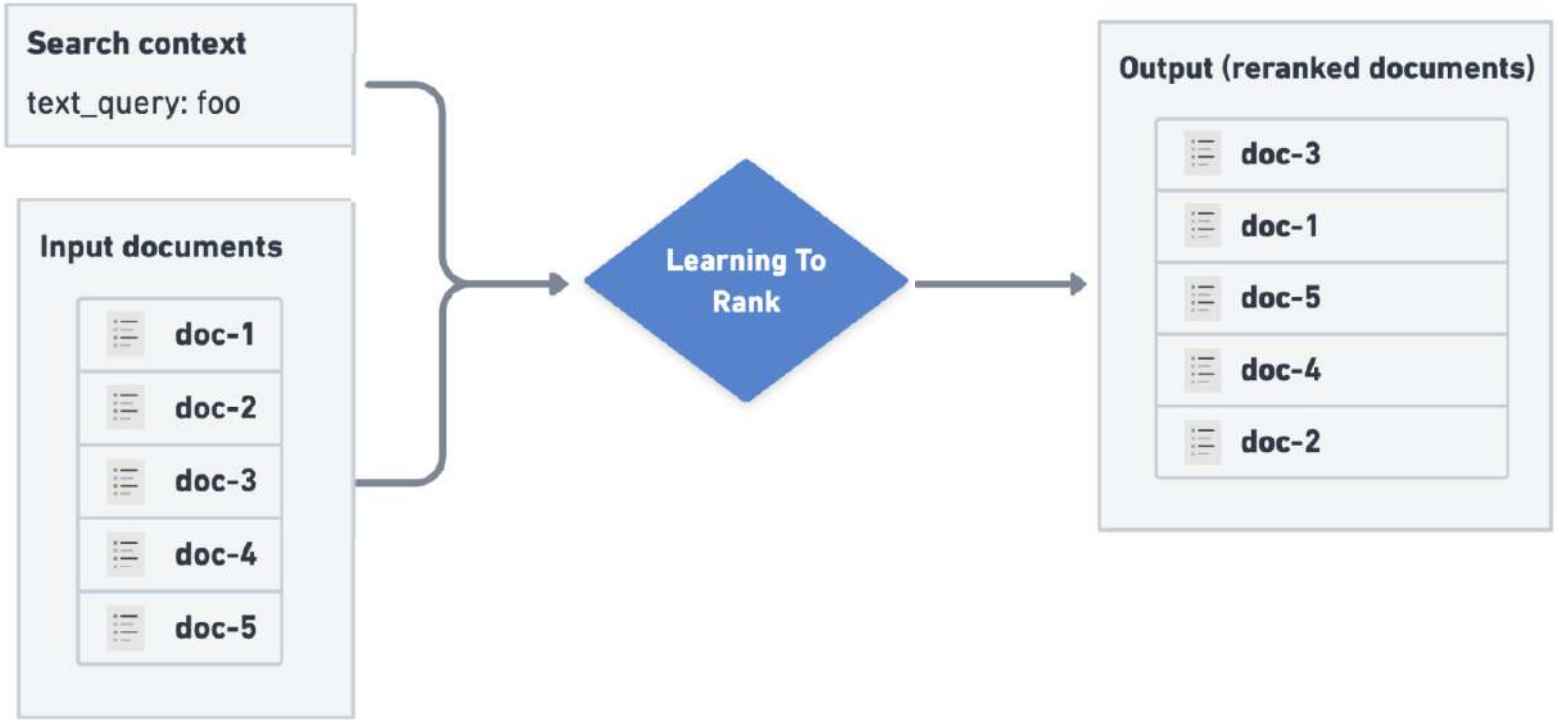

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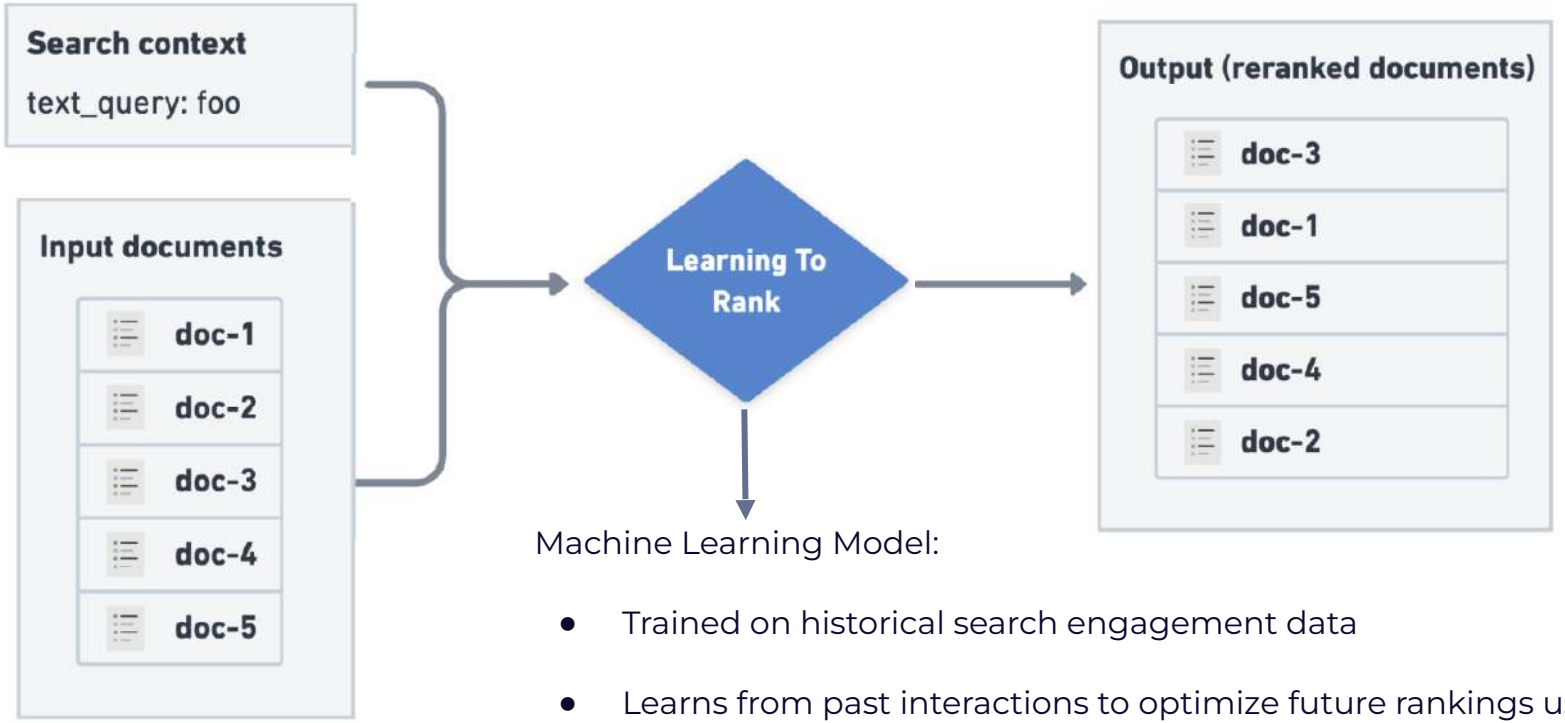

Machine Learning Model:

- Trained on historical search engagement data
- Learns from past interactions to optimize future rankings using a set of features

XGBoost LamdaMart

https://xgboost.readthedocs.io/en/latest/tutorials/learning_to_rank.html

DMLC XGBoost

Search docs

Installation Guide
Building From Source
Get Started with XGBoost
XGBoost Tutorials
Introduction to Boosted Trees
Introduction to Model IO
Slicing Models
Learning to Rank
DART booster
Monotonic Constraints
Feature Interaction Constraints
Survival Analysis with Accelerated Failure Time
Categorical Data
Multiple Outputs
Random Forests(TM) in XGBoost
Distributed XGBoost on Kubernetes
Distributed XGBoost with XGBoost4J-Spark
Distributed XGBoost with XGBoost4J-Spark-GPU

Learning to Rank

Contents

- Overview
- Training with the Pairwise Objective
- Position Bias
- Loss
- Constructing Pairs
- Obtaining Good Result
- Distributed Training
- Comparing Results with Version 1.7
- Reproducible Result
- References

Overview

Often in the context of information retrieval, learning-to-rank aims to train a model that arranges a set of query results into an ordered list [1]. For supervised learning-to-rank, the predictors are sample documents encoded as feature matrix, and the labels are relevance degree for each sample. Relevance degree can be multi-level (graded) or binary (relevant or not). The training samples are often grouped by their query index with each query group containing multiple query results.

XGBoost implements learning to rank through a set of objective functions and performance metrics. The default objective is `rank:ndcg` based on the `LambdaMART` [2] algorithm, which in turn is an adaptation of the `LambdaRank` [3] framework to gradient boosting trees. For a history and a summary of the algorithm, see [5]. The implementation in XGBoost features deterministic GPU computation, distributed training, position debiasing and two different pair construction strategies.

Open-source library

support for missing values

support for categorical features

Can use CPU and GPU

Fast in both training and inference speed

Different Modeling Approaches

By Judgment Granularity:

- Query-level (Coarse):

Aggregate engagement signals over a historical period

E.g. number of clicks or studies on a document in X days

- Session-level (Granular):

Use engagement signals with a single user session

E.g. whether the user clicked a document (0/1)

LambdaMart Explained

LamdaMart

LamdaMart

LamdaMart

LamdaMart

(1) Lambda:

Training target for each document

LamdaMart

(1) Lambda:

Training target for each document

LamdaMart

(1) Lambda:

Training target for each document

(2) MART:

The actual algorithm that learns from those targets

LamdaMart

(1) Lambda:

Training target for each document

Calculated using NDCG

(2) MART:

The actual algorithm that learns from those targets

LamdaMart

(1) Lambda:

Training target for each document

Calculated using NDCG

(2) MART:

The actual algorithm that learns from those targets

Gradient Boosting Regression Trees

(1) Calculate Lambdas

By Swapping

NDCG Quick Refresh

NDCG Quick Refresh

Discounted Gain (DG) = $$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

NDCG Quick Refresh

rel = 1, i = 1,

Discounted Gain (DG) = $$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

NDCG Quick Refresh

rel = 1, i = 1,

$$DG_1 = \frac{2^1 - 1}{\log_2(1+1)} = 1$$

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rel = 1, i = 3,

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Discounted Gain (DG) = $$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

rel = 1, i = 1,

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rel = 1, i = 3,

$$DG_3 = \frac{2^1 - 1}{\log_2(3+1)} = 0.5$$

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Discounted Cumulative Gain (DCG@k) = $\sum_{i=1}^{k}$ **Discounted Gain (DG)**

= **score of current ranking**

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Ideal DCG (IDCG@k) = score of ideal ranking (most relevant items at the top)

NDCG Quick Refresh

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rel = 1, i = 3,

$$DG_3 = \frac{2^1 - 1}{\log_2(3+1)} = 0.5$$

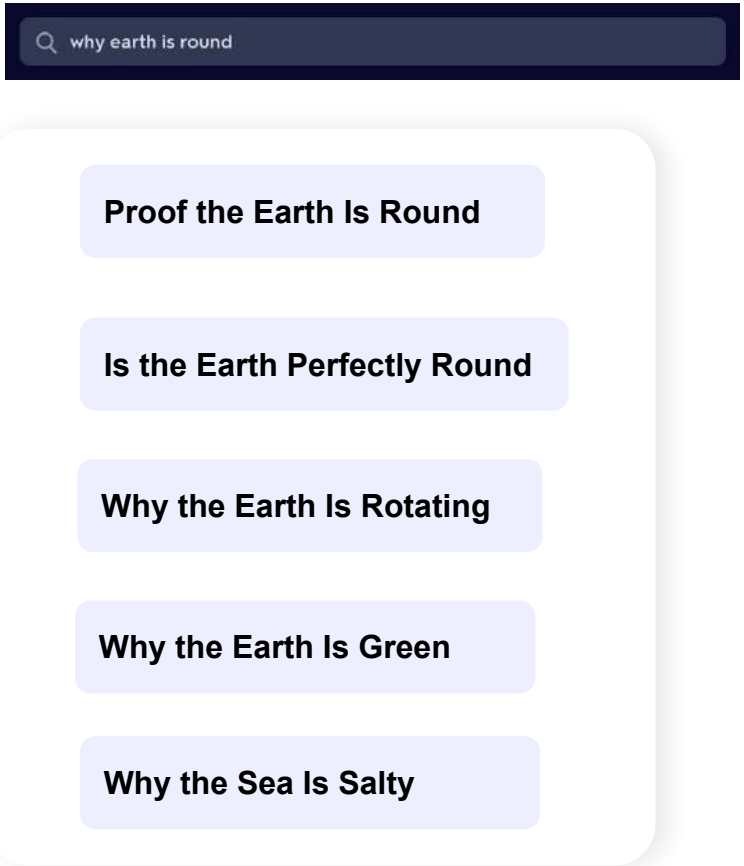
Discounted Cumulative Gain (DCG@k) = $\sum_{i=1}^{k}$ **Discounted Gain (DG)**

= **score of current ranking**

Ideal DCG (IDCG@k) = score of ideal ranking (most relevant items at the top)

$$NDCG@k = \frac{DCG@k}{IDCG@k}$$

Ideal Ranking



Studied Or Not (Relevance)
1
1
0
0
0

Ideal Ranking

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

Discounted Gain (DG)

why earth is round

	Studied Or Not (Relevance)
Proof the Earth Is Round	1
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Why the Sea Is Salty	0

Ideal Ranking

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Proof the Earth Is Round	1	1
Is the Earth Perfectly Round	1	
Why the Earth Is Rotating	0	
Why the Earth Is Green	0	
Why the Sea Is Salty	0	

Ideal Ranking

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Proof the Earth Is Round	1	1
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	
Why the Earth Is Green	0	
Why the Sea Is Salty	0	

Ideal Ranking

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

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why earth is round

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Is the Earth Perfectly Round

Why the Earth Is Rotating

Why the Earth Is Green

Why the Sea Is Salty

Studied Or Not (Relevance)		Discounted Gain (DG)
1		1
1	DCG = 1.631	0.631
0		0
0		0
0		0

Ideal Ranking

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

Proof the Earth Is Round

Is the Earth Perfectly Round

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Why the Sea Is Salty

Studied Or Not (Relevance)		Discounted Gain (DG)
1		1
1	DCG = 1.631	0.631
0	IDCG = 1.631	0
0		0
0		0

Ideal Ranking

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

Proof the Earth Is Round

Is the Earth Perfectly Round

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Why the Sea Is Salty

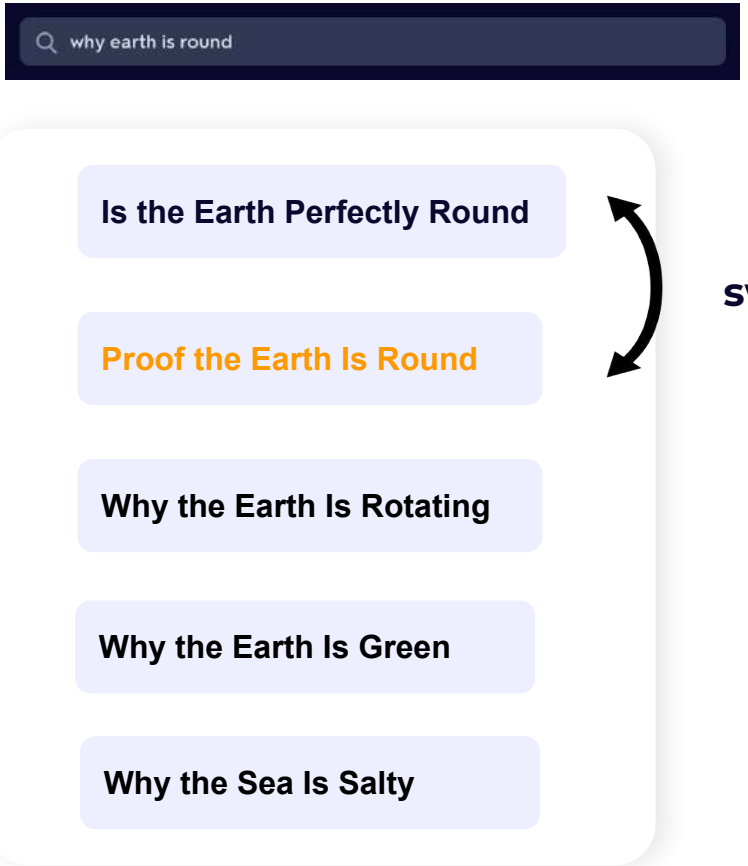
Studied Or Not (Relevance)		Discounted Gain (DG)
1		1
1	DCG = 1.631	0.631
0	IDCG = 1.631	0
	NDCG = 1	
0		0
0		0

Swapping 1

why earth is round

	Studied Or Not (Relevance)
Proof the Earth Is Round	1
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Why the Sea Is Salty	0

Swapping 1 and 2



swap

Studied Or Not (Relevance)
1
1
0
0
0

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

Discounted Gain (DG)

why earth is round

	Studied Or Not (Relevance)
Is the Earth Perfectly Round	1
Proof the Earth Is Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Why the Sea Is Salty	0

swap

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Is the Earth Perfectly Round	1	1
Proof the Earth Is Round	1	
Why the Earth Is Rotating	0	
Why the Earth Is Green	0	
Why the Sea Is Salty	0	

swap

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

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Swapping 1 and 2

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Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

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swap

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

swap

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Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Is the Earth Perfectly Round	1	1
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Why the Earth Is Green	0	0
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swap

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Is the Earth Perfectly Round	1	1
Proof the Earth Is Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Why the Sea Is Salty	0	0

swap

DCG = 1.631

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Is the Earth Perfectly Round	1	1
Proof the Earth Is Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Why the Sea Is Salty	0	0

swap

DCG = 1.631

IDCG = 1.631

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Is the Earth Perfectly Round	1	1
Proof the Earth Is Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
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swap

DCG = 1.631

IDCG = 1.631

NDCG = 1

Swapping 1 and 2

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Is the Earth Perfectly Round	1	1
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Why the Earth Is Green	0	0
Why the Sea Is Salty	0	0

swap

DCG = 1.631

IDCG = 1.631

NDCG = 1

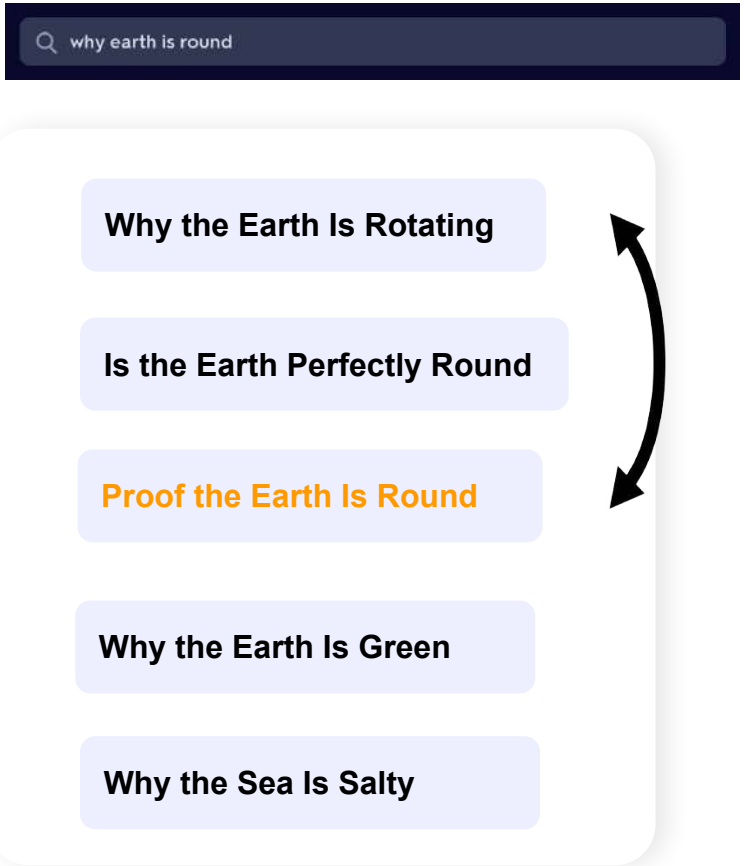
No Impact on NDCG!

Swapping 1

why earth is round

	Studied Or Not (Relevance)
Proof the Earth Is Round	1
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Why the Sea Is Salty	0

Swapping 1 and 3



Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

Discounted Gain (DG)

why earth is round

	Studied Or Not (Relevance)
Why the Earth Is Rotating	0
Is the Earth Perfectly Round	1
Proof the Earth Is Round	1
Why the Earth Is Green	0
Why the Sea Is Salty	0

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	
Proof the Earth Is Round	1	
Why the Earth Is Green	0	
Why the Sea Is Salty	0	

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	0.631
Proof the Earth Is Round	1	
Why the Earth Is Green	0	
Why the Sea Is Salty	0	

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	0.631
Proof the Earth Is Round	1	0.5
Why the Earth Is Green	0	
Why the Sea Is Salty	0	

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	0.631
Proof the Earth Is Round	1	0.5
Why the Earth Is Green	0	0
Why the Sea Is Salty	0	

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	0.631
Proof the Earth Is Round	1	0.5
Why the Earth Is Green	0	0
Why the Sea Is Salty	0	0

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	0.631
Proof the Earth Is Round	1	0.5
Why the Earth Is Green	0	0
Why the Sea Is Salty	0	0

swap

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

Why the Earth Is Rotating

Is the Earth Perfectly Round

Proof the Earth Is Round

Why the Earth Is Green

Why the Sea Is Salty

swap

Studied Or Not (Relevance)	Discounted Gain (DG)
0	0
1	0.631
1	0.5
0	0
0	0

DCG = 1.131

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Rotating	0	0
Is the Earth Perfectly Round	1	0.631
Proof the Earth Is Round	1	0.5
Why the Earth Is Green	0	0
Why the Sea Is Salty	0	0

swap

DCG = 1.131

IDCG = 1.631

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

Why the Earth Is Rotating

Is the Earth Perfectly Round

Proof the Earth Is Round

Why the Earth Is Green

Why the Sea Is Salty

swap

Studied Or Not (Relevance)	Discounted Gain (DG)
0	0
1	0.631
1	0.5
0	0
0	0

DCG = 1.131

IDCG = 1.631

NDCG = 0.693

Swapping 1 and 3

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

Why the Earth Is Rotating

Is the Earth Perfectly Round

Proof the Earth Is Round

Why the Earth Is Green

Why the Sea Is Salty

swap

Studied Or Not (Relevance)	Discounted Gain (DG)
0	0
1	0.631
1	0.5
0	0
0	0

DCG = 1.131

IDCG = 1.631

NDCG = 0.693

NDCG 0.307 ↓

Swapping 1

why earth is round

	Studied Or Not (Relevance)
Proof the Earth Is Round	1
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Why the Sea Is Salty	0

Swapping 1 and 4

Why the Earth Is Green

Is the Earth Perfectly Round

Why the Earth Is Rotating

Proof the Earth Is Round

Why the Sea Is Salty

swap

Studied Or Not

(Relevance)

0

1

0

1

0

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

Discounted Gain (DG)

why earth is round

	Studied Or Not (Relevance)
Why the Earth Is Green	0
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Proof the Earth Is Round	1
Why the Sea Is Salty	0

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	
Why the Earth Is Rotating	0	
Proof the Earth Is Round	1	
Why the Sea Is Salty	0	

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	
Proof the Earth Is Round	1	
Why the Sea Is Salty	0	

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	
Why the Sea Is Salty	0	

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	0.431
Why the Sea Is Salty	0	

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	0.431
Why the Sea Is Salty	0	0

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	0.431
Why the Sea Is Salty	0	0

swap

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	0.431
Why the Sea Is Salty	0	0

swap

DCG = 1.062

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	0.431
Why the Sea Is Salty	0	0

swap

DCG = 1.062

IDCG = 1.631

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Earth Is Green	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Proof the Earth Is Round	1	0.431
Why the Sea Is Salty	0	0

swap

DCG = 1.062

IDCG = 1.631

NDCG = 0.651

Swapping 1 and 4

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

Why the Earth Is Green

Is the Earth Perfectly Round

Why the Earth Is Rotating

Proof the Earth Is Round

Why the Sea Is Salty

swap

Studied Or Not (Relevance)	Discounted Gain (DG)
0	0
1	0.631
0	0
1	0.431
0	0

DCG = 1.062

IDCG = 1.631

NDCG = 0.651

NDCG 0.349 ↓

Swapping 1

why earth is round

	Studied Or Not (Relevance)
Proof the Earth Is Round	1
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Why the Sea Is Salty	0

Swapping 1 and 5

Why the Sea Is Salty

Is the Earth Perfectly Round

Why the Earth Is Rotating

Why the Earth Is Green

Proof the Earth Is Round

swap

Studied Or Not

(Relevance)

0

1

0

0

1

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

Discounted Gain (DG)

why earth is round

	Studied Or Not (Relevance)
Why the Sea Is Salty	0
Is the Earth Perfectly Round	1
Why the Earth Is Rotating	0
Why the Earth Is Green	0
Proof the Earth Is Round	1

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	
Why the Earth Is Rotating	0	
Why the Earth Is Green	0	
Proof the Earth Is Round	1	

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	
Why the Earth Is Green	0	
Proof the Earth Is Round	1	

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	
Proof the Earth Is Round	1	

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	0.387

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	0.387

swap

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	0.387

swap

DCG = 1.018

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	0.387

swap

DCG = 1.018

IDCG = 1.631

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	0.387

swap

DCG = 1.018

IDCG = 1.631

NDCG = 0.624

Swapping 1 and 5

$$\frac{2^{rel_i} - 1}{\log_2(i+1)}$$

why earth is round

	Studied Or Not (Relevance)	Discounted Gain (DG)
Why the Sea Is Salty	0	0
Is the Earth Perfectly Round	1	0.631
Why the Earth Is Rotating	0	0
Why the Earth Is Green	0	0
Proof the Earth Is Round	1	0.387

swap

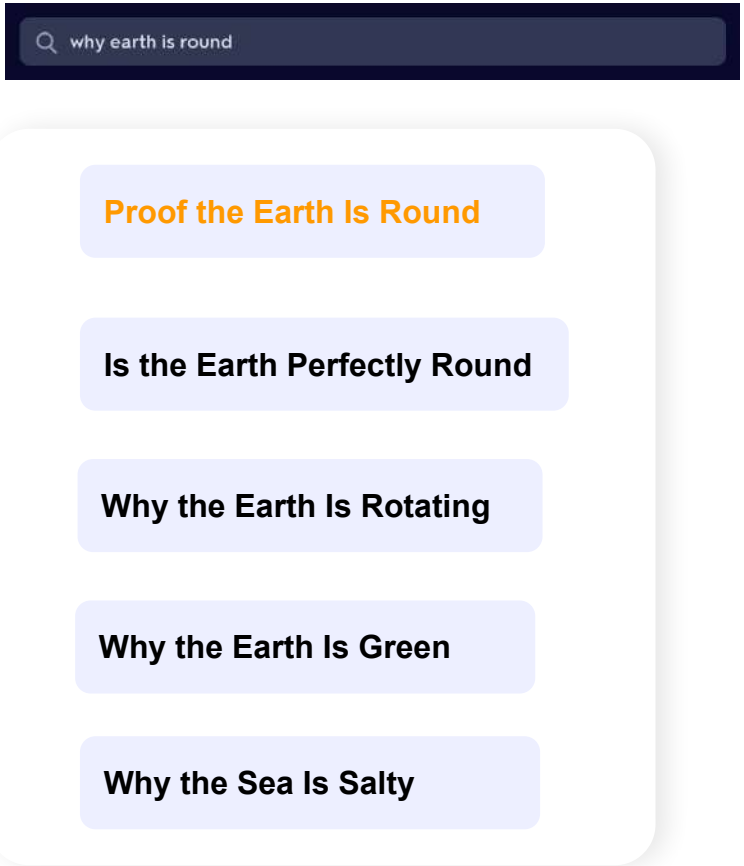
DCG = 1.018

IDCG = 1.631

NDCG = 0.624

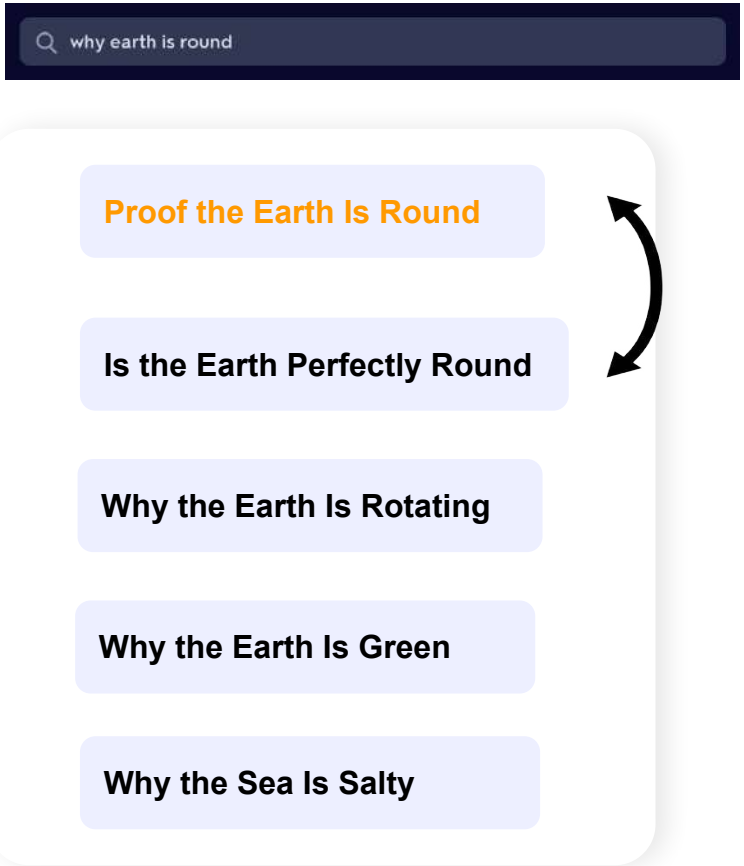
NDCG 0.376 ↓

Calculate Lambdas



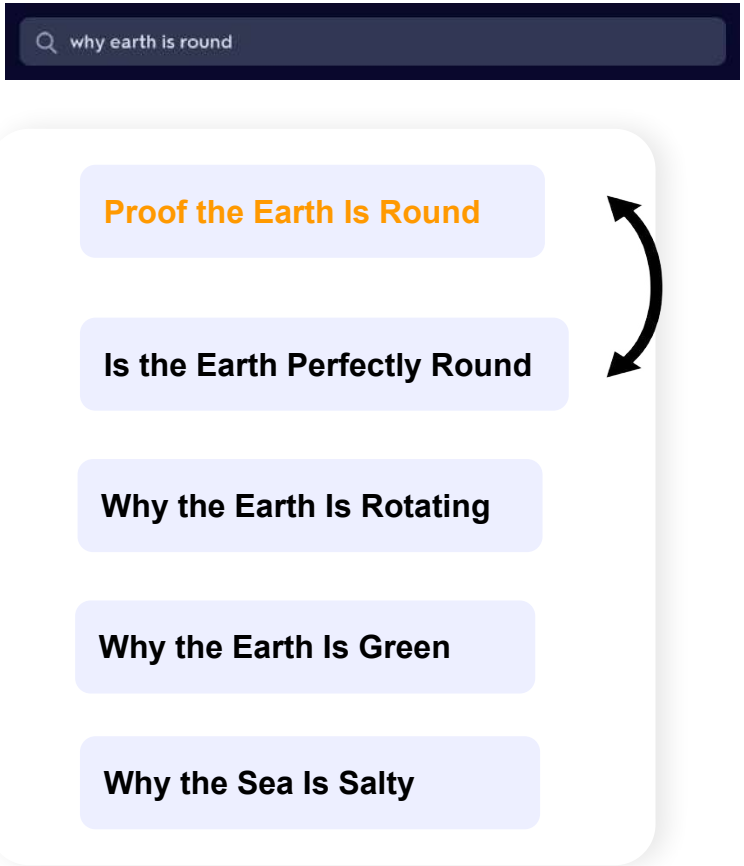
Lambdas

Calculate Lambdas



Lambdas

Calculate Lambdas

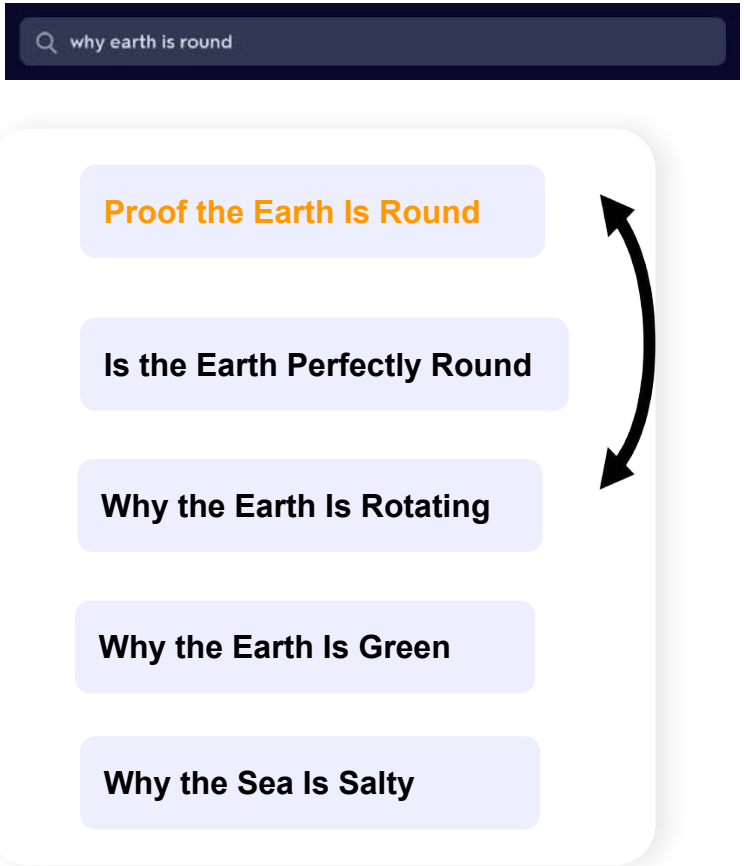


Lambdas

0

0

Calculate Lambdas

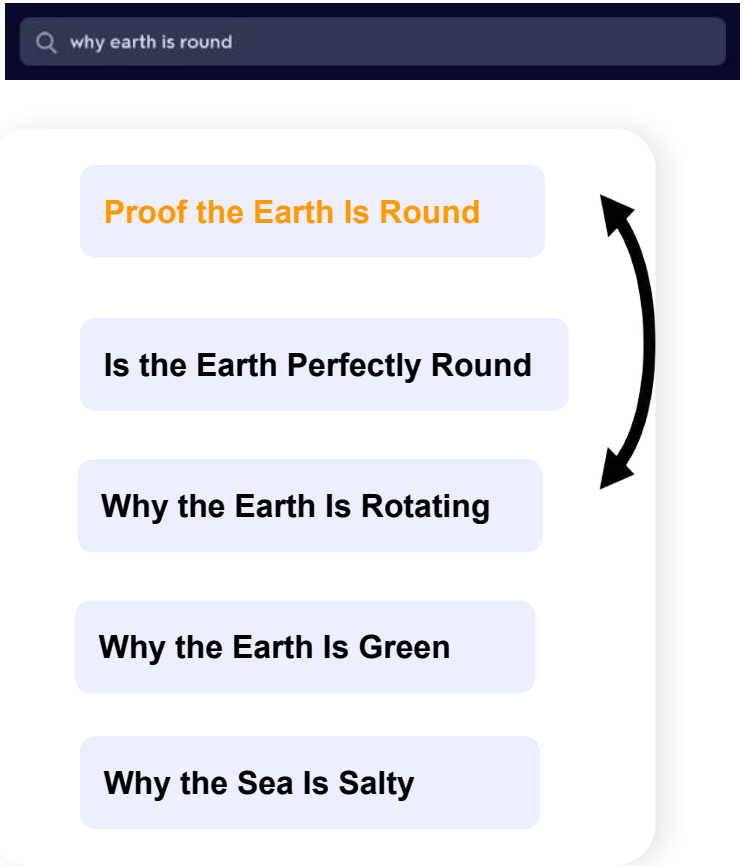


Lambdas

0

0

Calculate Lambdas



Lambdas

0 + 0.307

0

-0.307

Calculate Lambdas

why earth is round

Proof the Earth Is Round

Is the Earth Perfectly Round

Why the Earth Is Rotating

Why the Earth Is Green

Why the Sea Is Salty

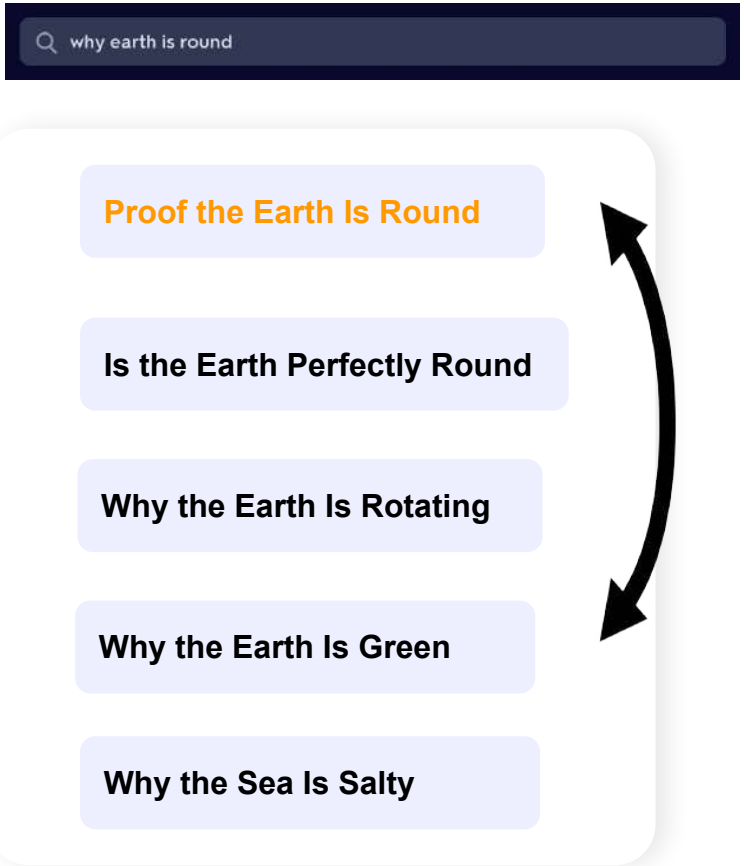
Lambdas

0 + 0.307

0

-0.307

Calculate Lambdas



Lambdas

0 + 0.307 + 0.349

0

-0.307

-0.349

Calculate Lambdas

Lambdas

0 + 0.307 + 0.349

0

-0.307

-0.349

Calculate Lambdas

Lambdas

0 + 0.307 + 0.349 + 0.376

0

-0.307

-0.349

-0.376

Calculate Lambdas

Lambdas

0 + 0.307 + 0.349 + 0.376 = 1.032

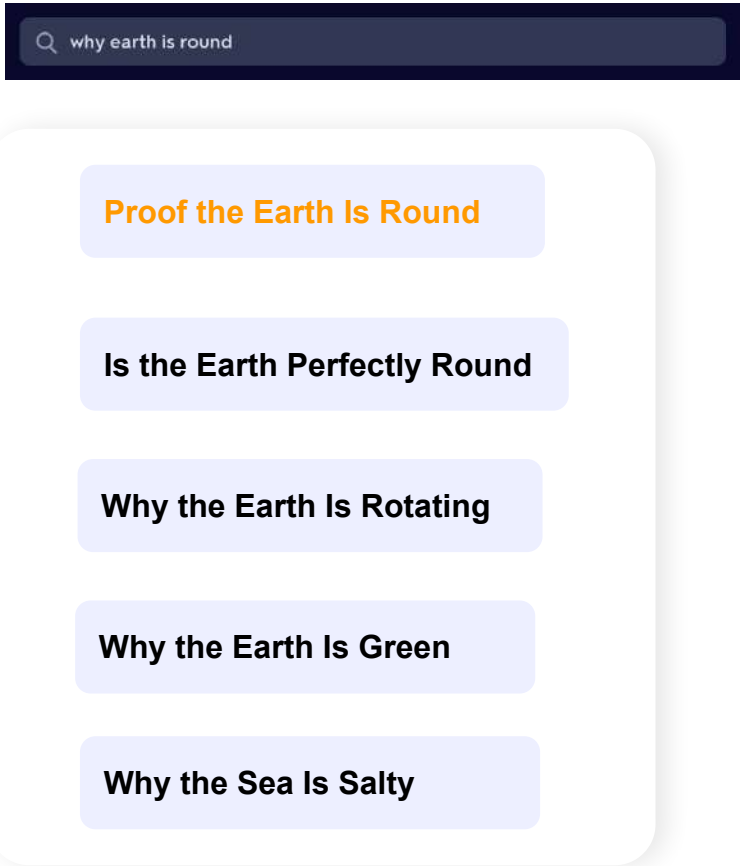
0

-0.307

-0.349

-0.376

Calculate Lambdas



Lambdas

1.032

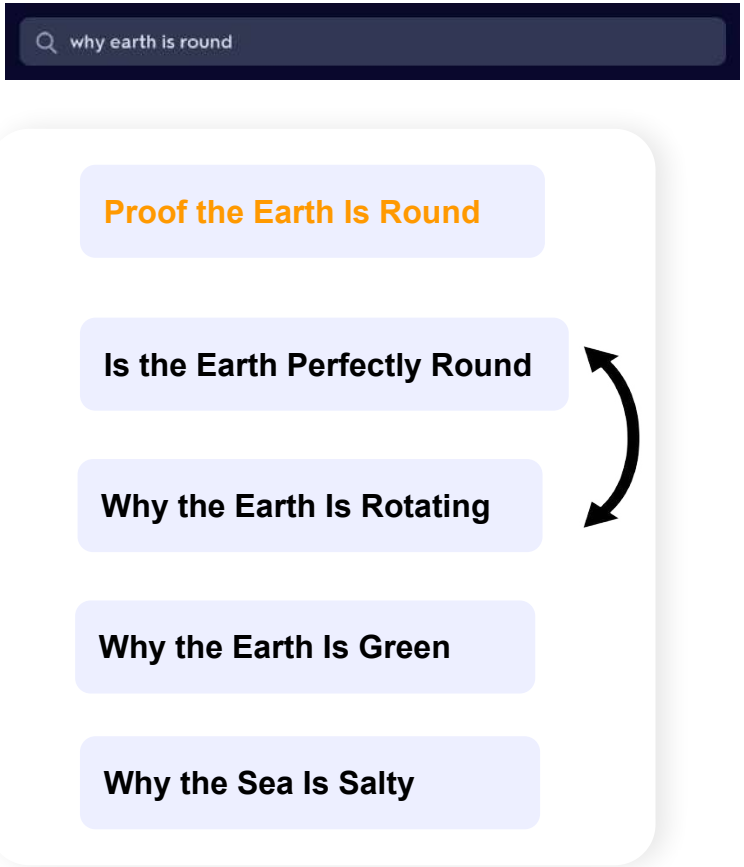
0

-0.307

-0.349

-0.376

Calculate Lambdas



Lambdas

1.032

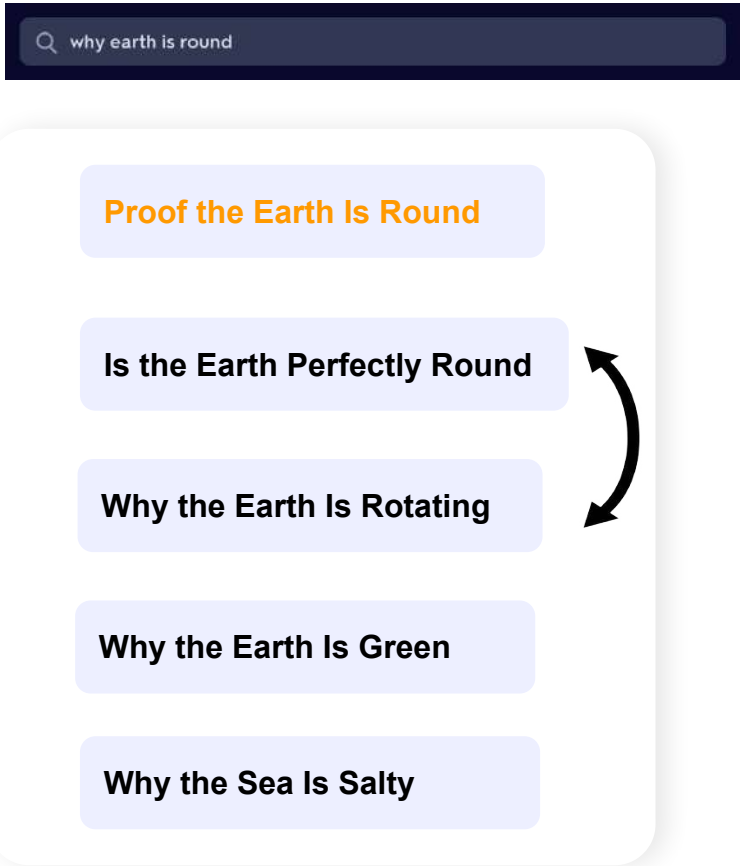
0

-0.307

-0.349

-0.376

Calculate Lambdas



Lambdas

1.032

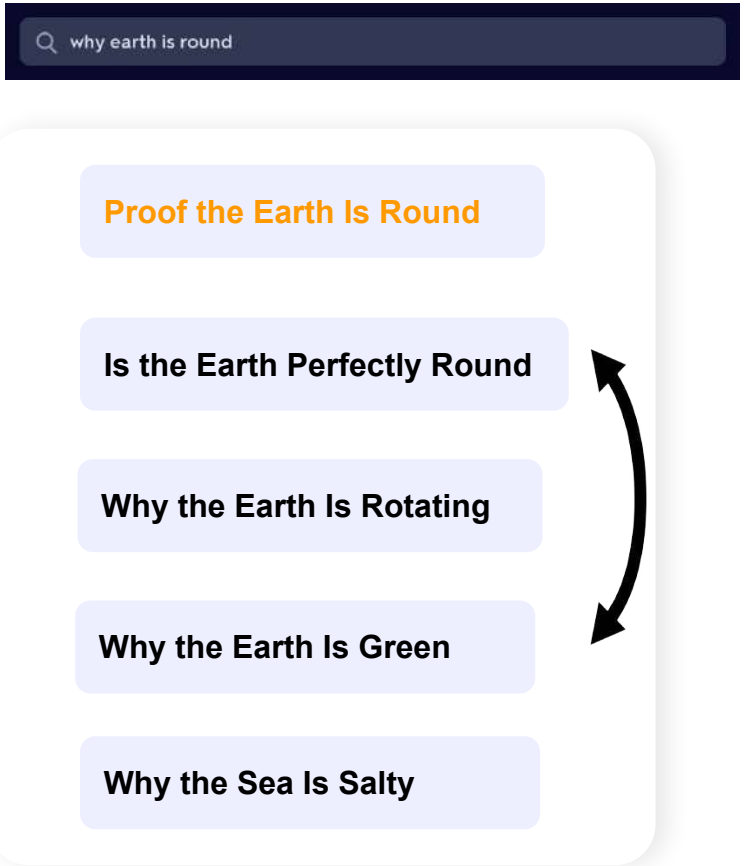
0 +0.080

-0.307 -0.080

-0.349

-0.376

Calculate Lambdas



Lambdas

1.032

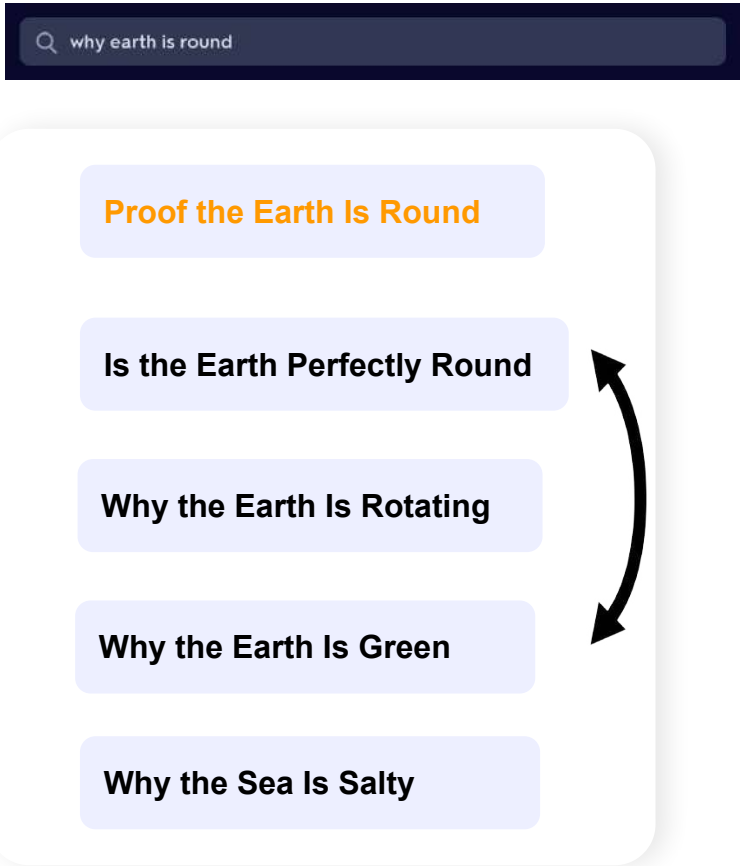
0 +0.080

-0.307 -0.080

-0.349

-0.376

Calculate Lambdas



Lambdas

1.032

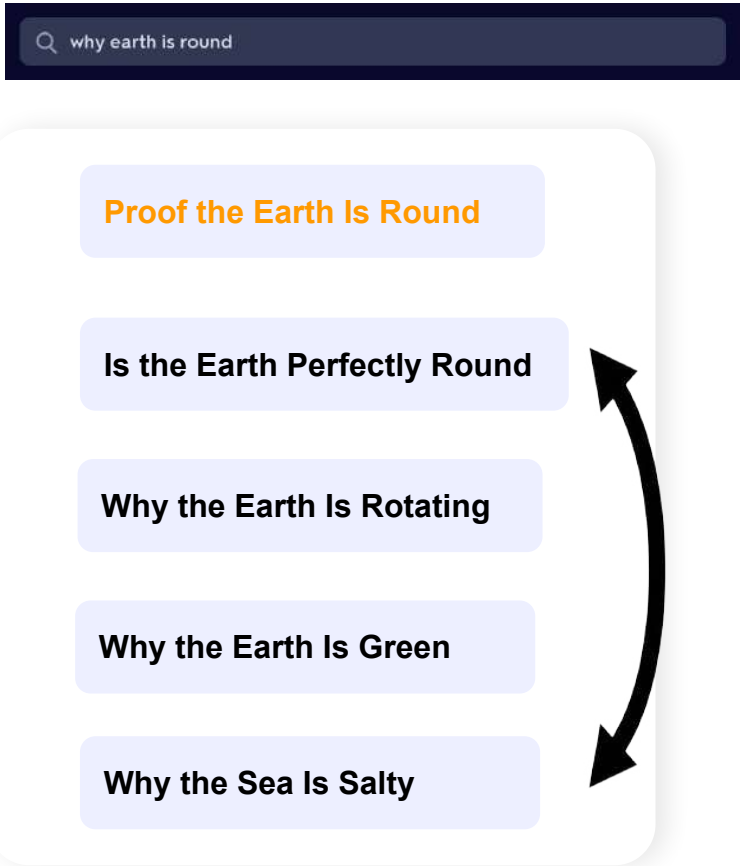
0 +0.080 +0.123

-0.307 -0.080

-0.349 -0.123

-0.376

Calculate Lambdas



Lambdas

1.032

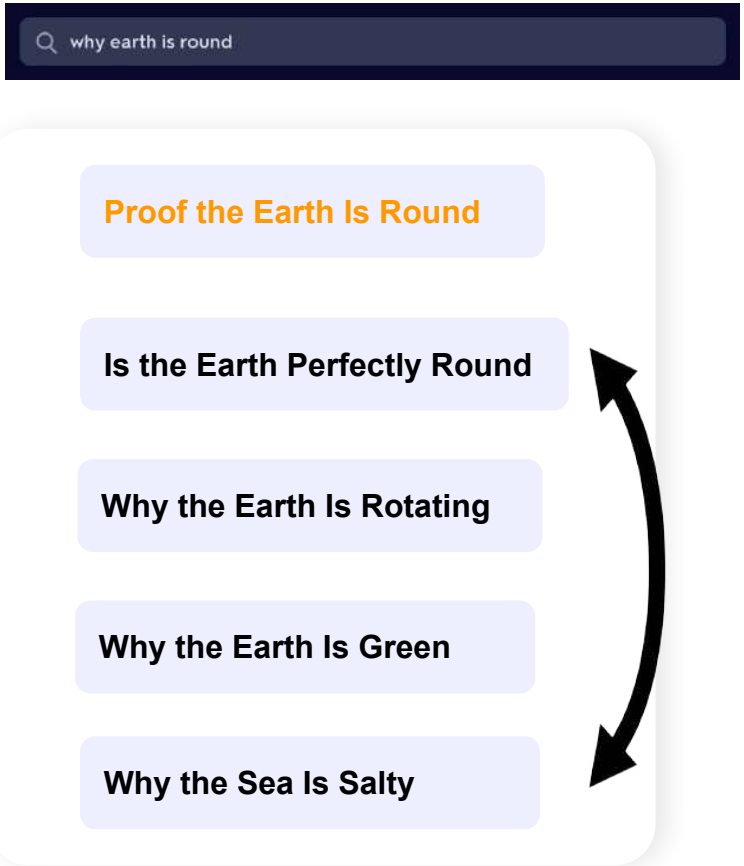
0 +0.080 +0.123

-0.307 -0.080

-0.349 -0.123

-0.376

Calculate Lambdas



Lambdas

1.032

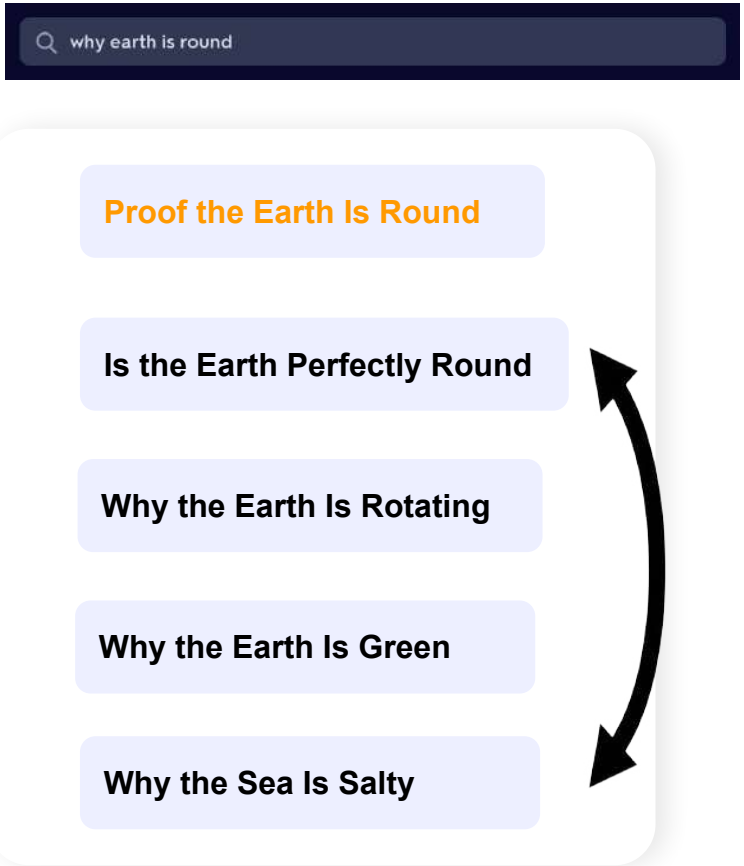
0 +0.080 +0.123 +0.150

-0.307 -0.080

-0.349 -0.123

-0.376 -0.150

Calculate Lambdas



Lambdas

1.032

0 +0.080 +0.123 +0.150 = 0.353

-0.307 -0.080 = -0.387

-0.349 -0.123 = -0.472

-0.376 -0.150 = -0.526

Calculate Lambdas

why earth is round

	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387
Why the Earth Is Green	-0.472
Why the Sea Is Salty	-0.526

Calculate Lambdas

why earth is round

	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387
Why the Earth Is Green	-0.472
Why the Sea Is Salty	-0.526

Calculate Lambdas

why earth is round

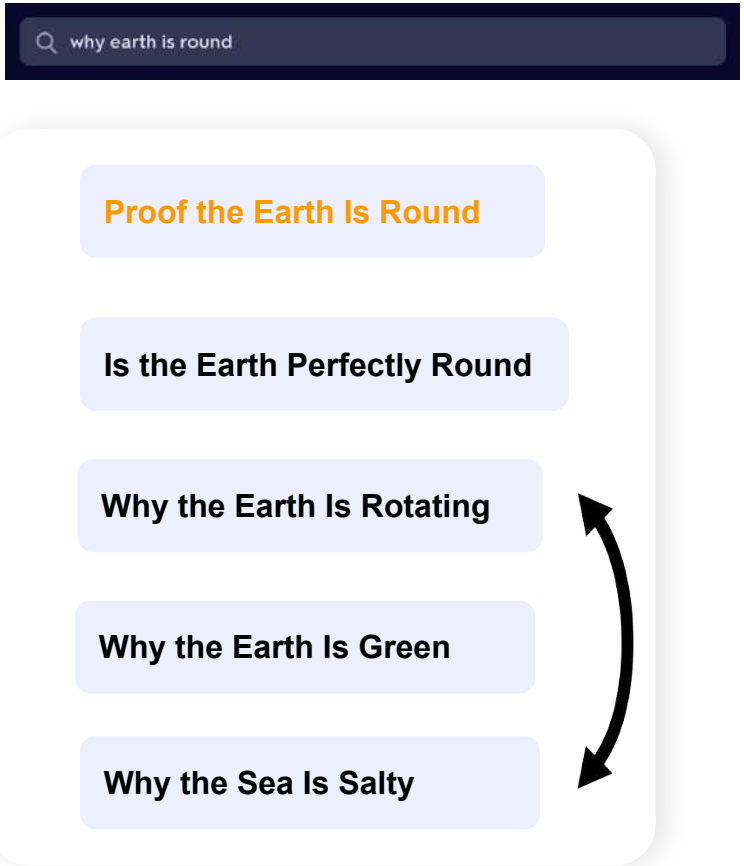
	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387 +0
Why the Earth Is Green	-0.472 -0
Why the Sea Is Salty	-0.526

Calculate Lambdas

why earth is round

	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387 +0
Why the Earth Is Green	-0.472 -0
Why the Sea Is Salty	-0.526

Calculate Lambdas



Lambdas

1.032

0.353

-0.387 +0 +0

-0.472 -0

-0.526 -0

Calculate Lambdas

why earth is round

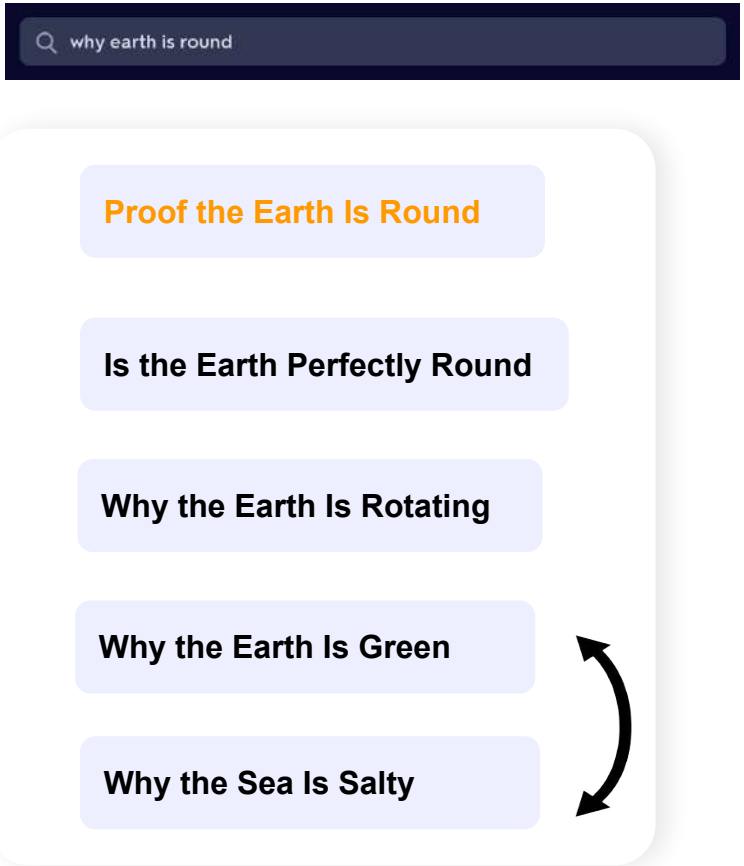
	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387
Why the Earth Is Green	-0.472
Why the Sea Is Salty	-0.526

Calculate Lambdas

why earth is round

	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387
Why the Earth Is Green	-0.472
Why the Sea Is Salty	-0.526

Calculate Lambdas



Lambdas

1.032

0.353

-0.387

-0.472 +0

-0.526 -0

Calculate Lambdas

why earth is round

	Lambdas
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387
Why the Earth Is Green	-0.472
Why the Sea Is Salty	-0.526

Calculate Lambdas

why earth is round

	Prediction Target
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0.353
Why the Earth Is Rotating	-0.387
Why the Earth Is Green	-0.472
Why the Sea Is Salty	-0.526

(2) MART

Multiple Additive Regression Trees (MART)

	Prediction Target
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0
Why the Earth Is Rotating	-0.307
Why the Earth Is Green	-0.349
Why the Sea Is Salty	-0.376

Multiple Additive Regression Trees (MART)

why earth is round

	Prediction Target
Proof the Earth Is Round	1.032
Is the Earth Perfectly Round	0
Why the Earth Is Rotating	-0.307
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Why the Sea Is Salty	-0.376

Multiple Additive Regression Trees (MART)

Document A features	Prediction Target
Is the Earth Perfectly Round	1.032
Why the Earth Is Rotating	0
Why the Earth Is Green	-0.307
Why the Sea Is Salty	-0.349
	-0.376

Multiple Additive Regression Trees (MART)

	Prediction Target
Document A features	1.032
Document B features	0
Why the Earth Is Rotating	-0.307
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Multiple Additive Regression Trees (MART)

	Prediction Target
Document A features	1.032
Document B features	0
Document C features	-0.307
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Multiple Additive Regression Trees (MART)

why earth is round

	Prediction Target
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Multiple Additive Regression Trees (MART)

why earth is round

	Prediction Target
Document A features	1.032
Document B features	0
Document C features	-0.307
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Document E features	-0.376

Multiple Additive Regression Trees (MART)

why earth is round

Document A features

Document B features

Document C features

Document D features

Document E features

Gradient Boosting Trees

Prediction Target

1.032

0

-0.307

-0.349

-0.376

Multiple Additive Regression Trees (MART)

Document A features

Document B features

Document C features

Document D features

Document E features

Gradient Boosting

Trees

Prediction Target

1.032

0

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-0.349

-0.376

Illustration Of A Gradient Boosting Tree

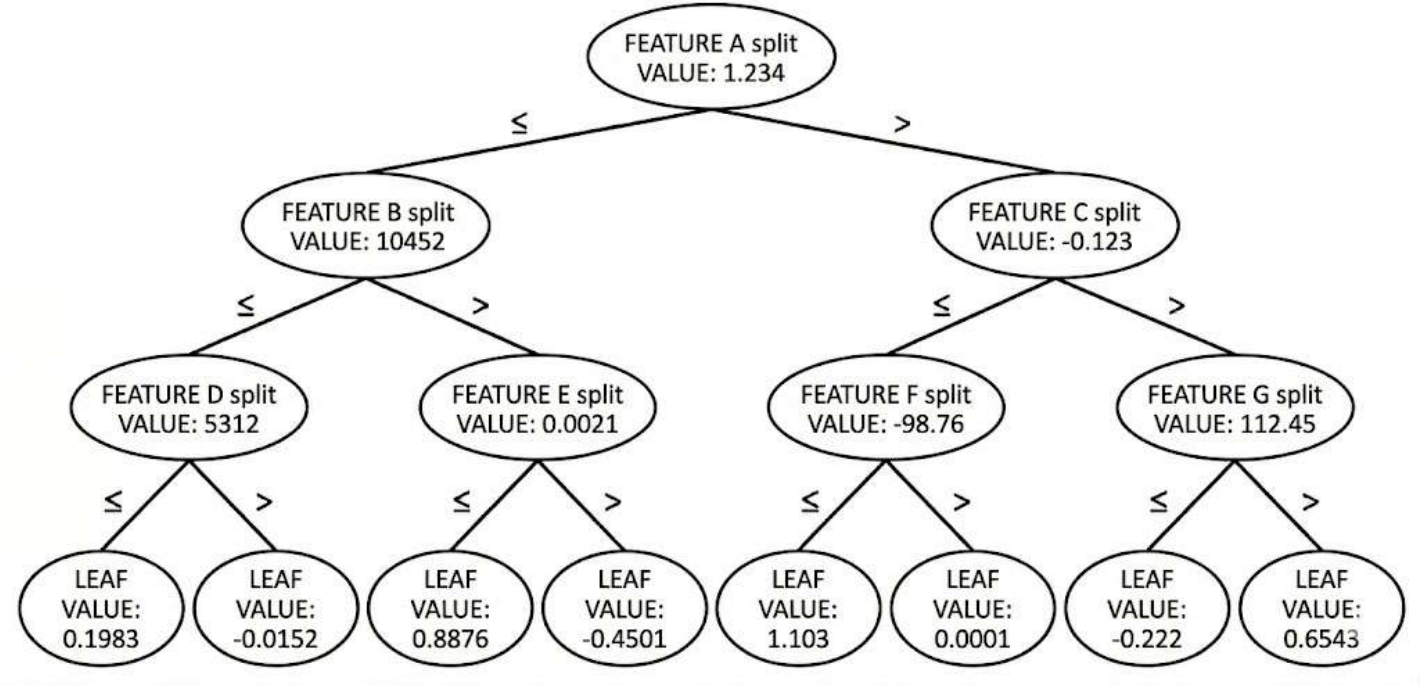
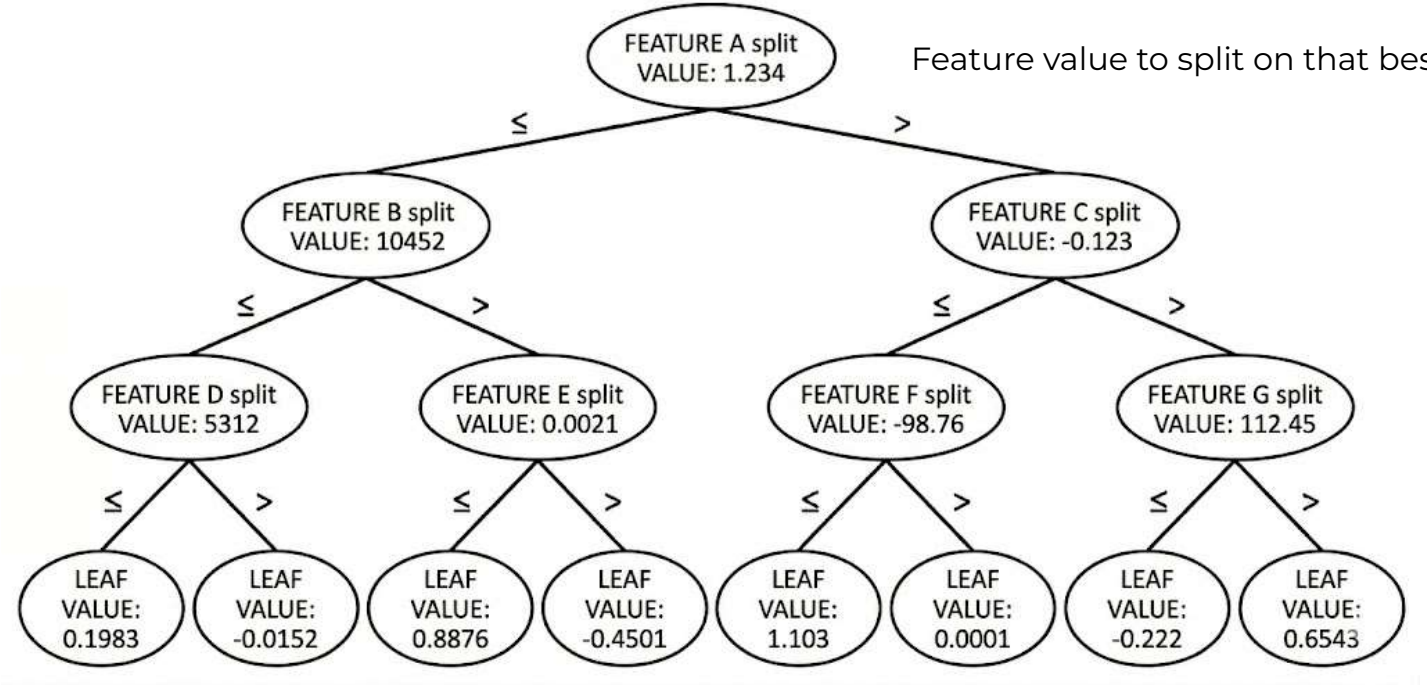


Illustration Of A Gradient Boosting Tree

(1) Learns the tree splits (decision rules)

Feature value to split on that best divide the data



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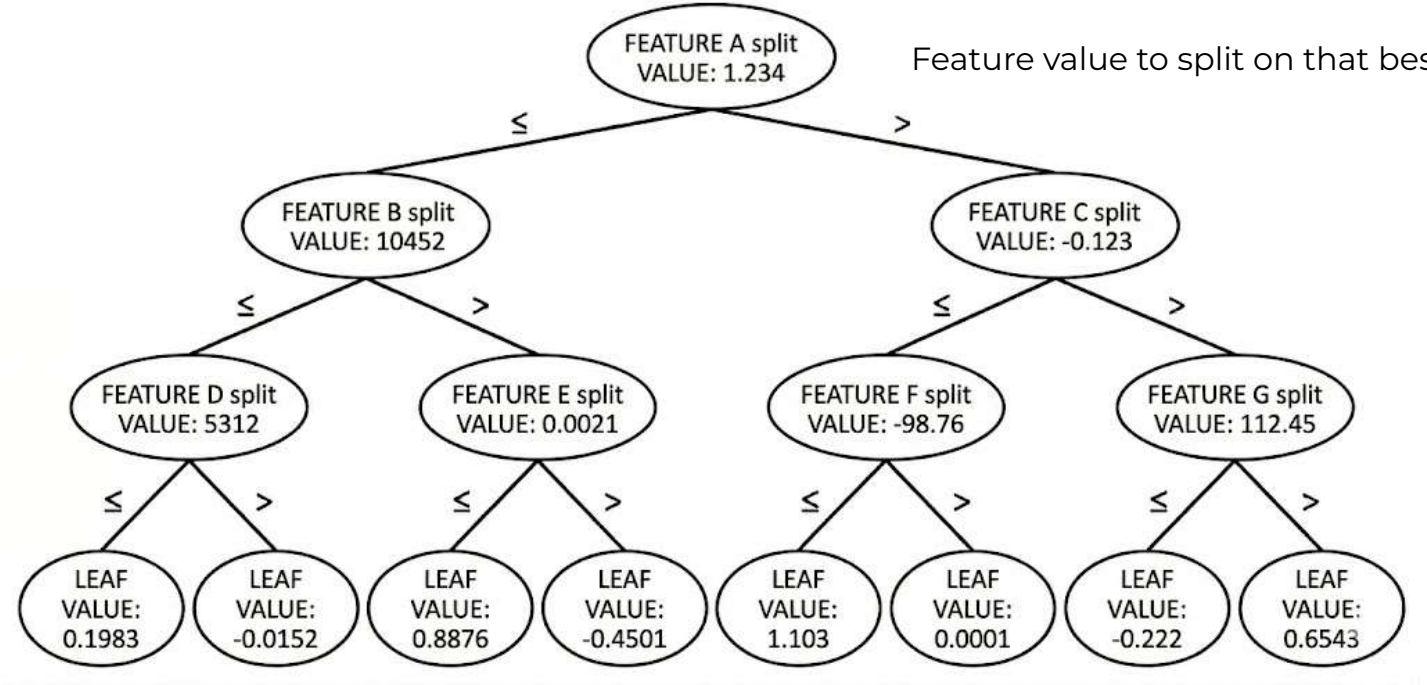
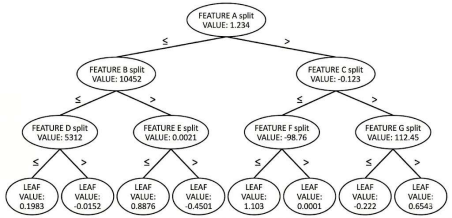
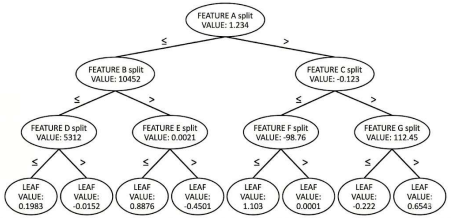


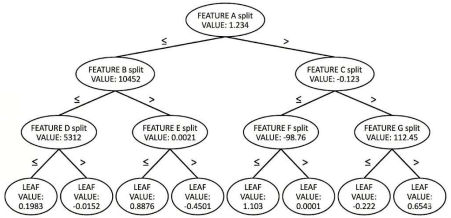
(2) Learns the leaf values (approximate the lambdas)

Gradient Boosting Trees

One tree is not enough







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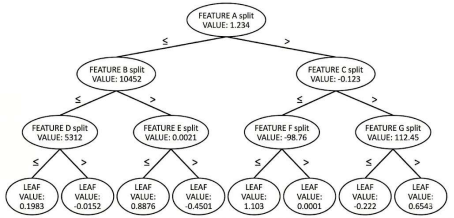

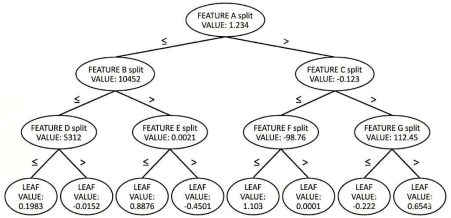

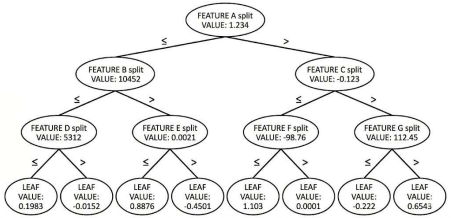

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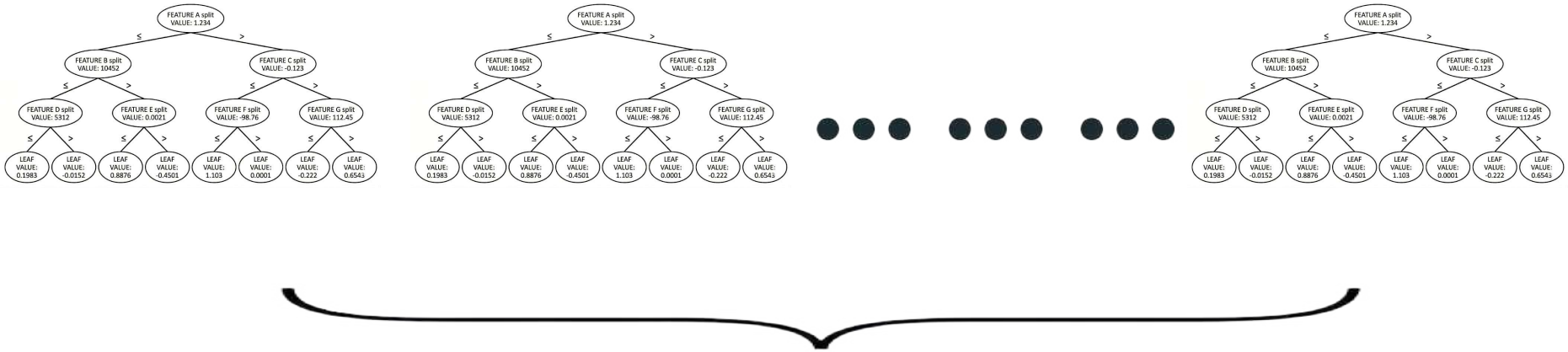

Number of Trees - a hyperparameter

XGBoost Implementation

https://xgboost.readthedocs.io/en/latest/python/examples/learning_to_rank.html

Search docs

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GPU Support

XGBoost Parameters

Prediction

Tree Methods

Python Package


```python
ranker = xgb.XGBRanker(
    n_estimators=512,
    tree_method="hist",
    device="cuda",
    learning_rate=0.01,
    reg_lambda=1.5,
    subsample=0.8,
    sampling_method="gradient_based",
    # LTR specific parameters
    objective="rank:ndcg",
    # - Enable bias estimation
    lambdarank_unbiased=True,
    # - normalization (1 / (norm + 1))
    lambdarank_bias_norm=1,
    # - Focus on the top 12 documents
    lambdarank_num_pair_per_sample=12,
    lambdarank_pair_method="topk",
    ndcg_exp_gain=True,
    eval_metric=["ndcg@1", "ndcg@3", "ndcg@5", "ndcg@10"],
    callbacks=[ShowPosition()],
)
ranker.fit(
    X_train,
    clicks_train,
    qid=qid_train,
    eval_set=[(X_test, y_test), (X_test, clicks_test)],
    eval_qid=[qid_test, qid_test],
    verbose=True,
)
ranker.predict(X_test)
```

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```

Deployment

Approach I: Microservice

Pros:

Avoid bloating Elastic Index

Lower cost

Iterate independently

Cons:

Extra dev work

Extra network round trips

Higher Latency

Approach II: Run As a Rescore Query

https://www.elastic.co/docs/solutions/search/ranking/learning-to-rank-search-usage

```
GET my-index/_search
{
  "query": { 1
    "multi_match": {
      "fields": ["title", "content"],
      "query": "the quick brown fox"
    }
  },
  "rescore": {
    "learning_to_rank": {
      "model_id": "ltr-model", 2
      "params": { 3
        "query_text": "the quick brown fox"
      }
    },
    "window_size": 100 4
  }
}
```

Pros:

No extra dev work outside Elastic

No extra network round trips

Low latency

Cons:

Bigger index size

Higher cost

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```

Pros:

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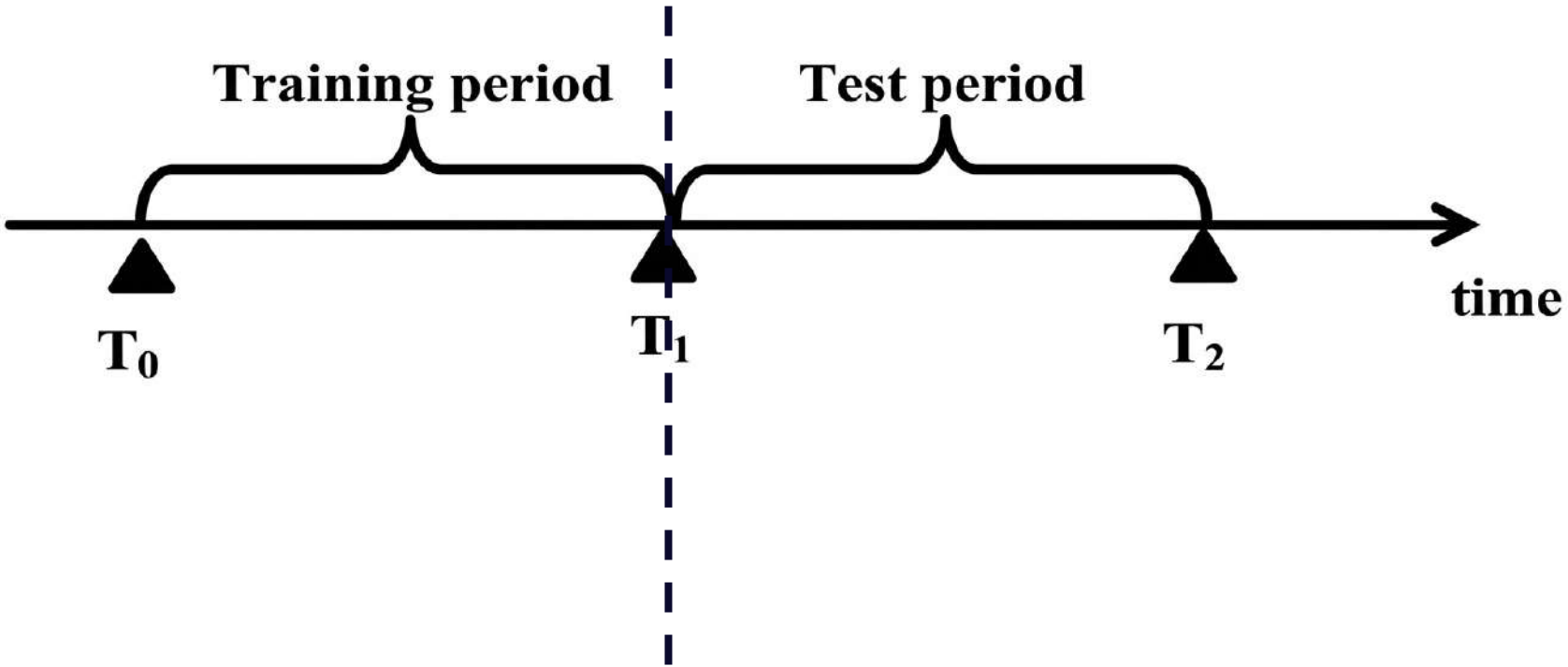
Offline Evaluation

Train Test temporal split



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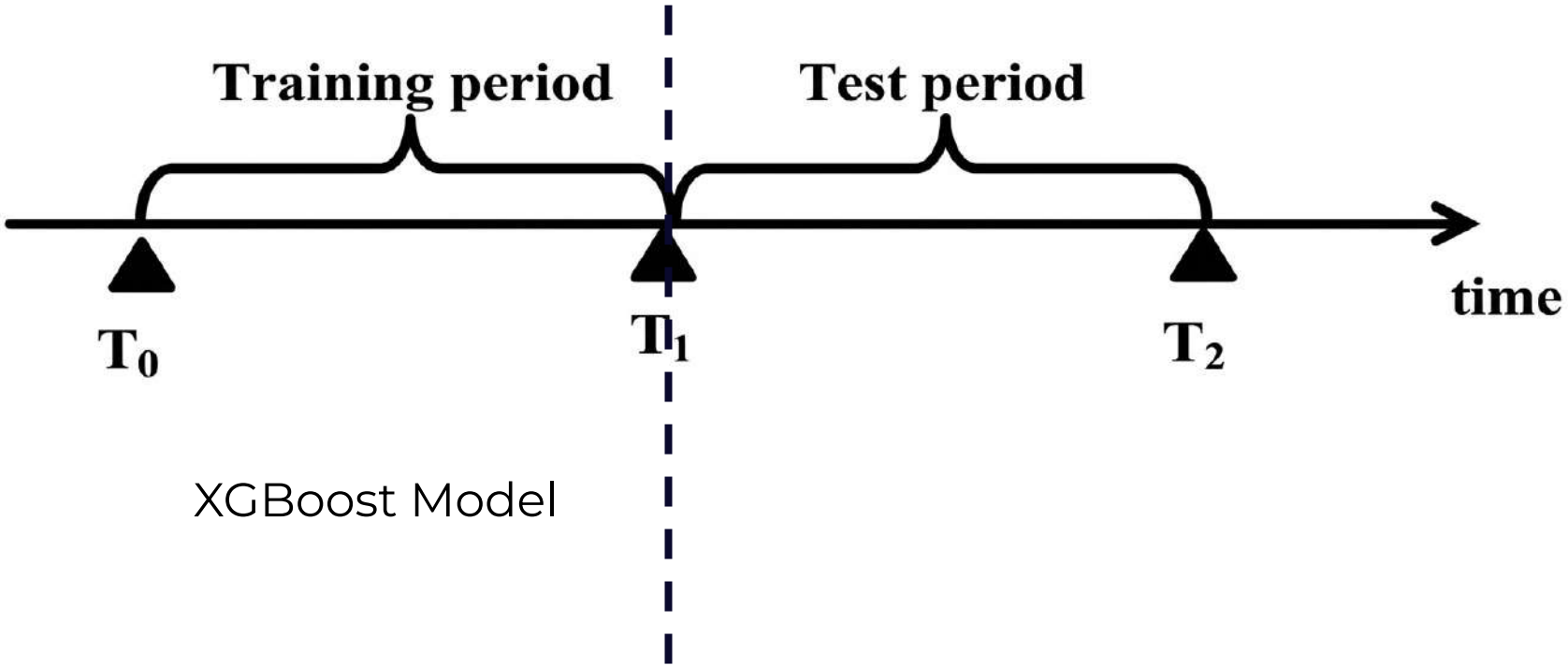
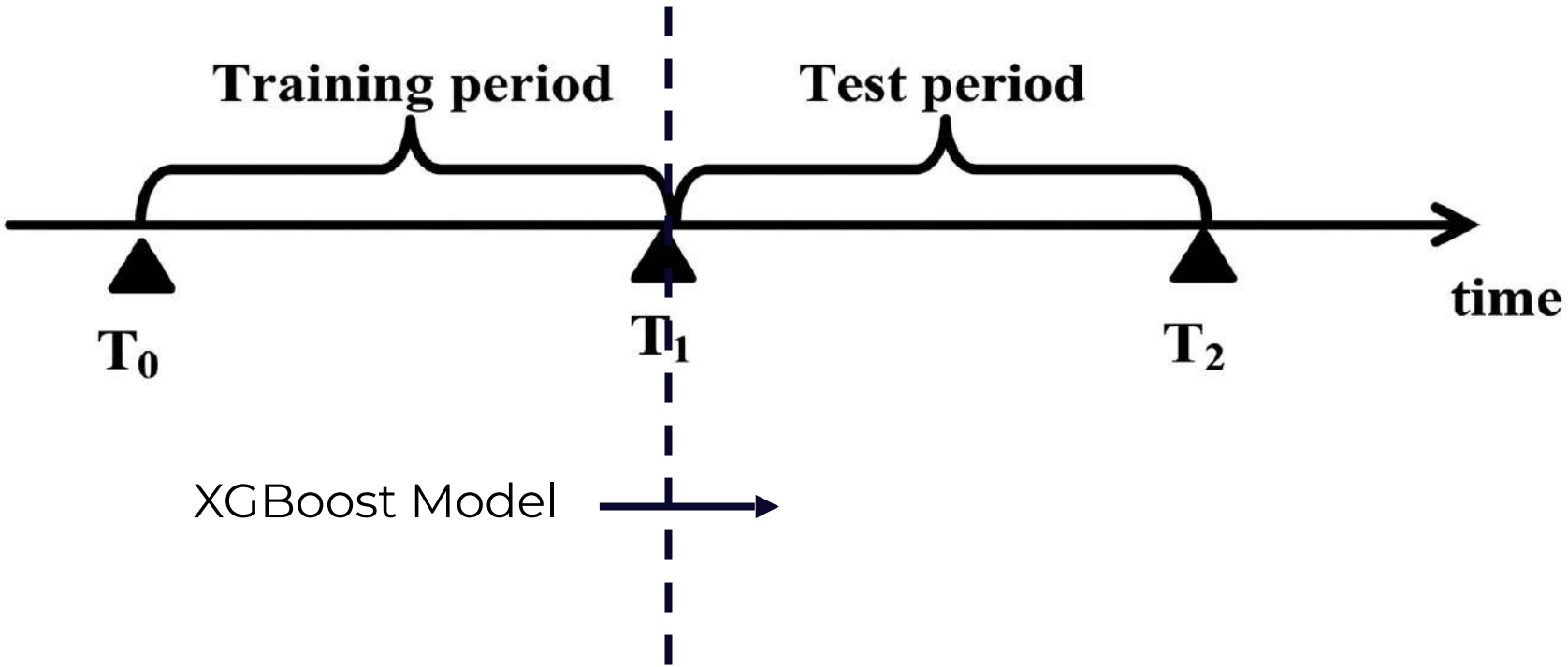


XGBoost Model

Train Test temporal split



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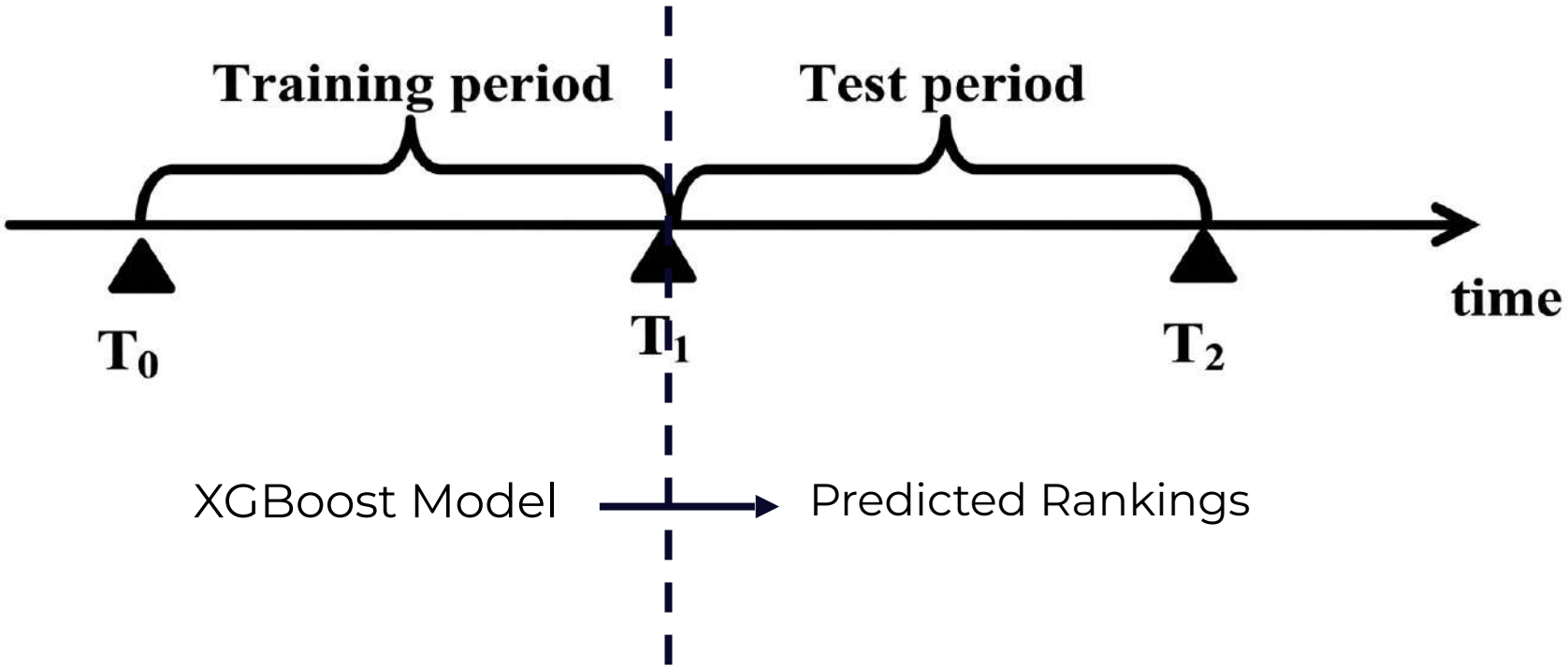

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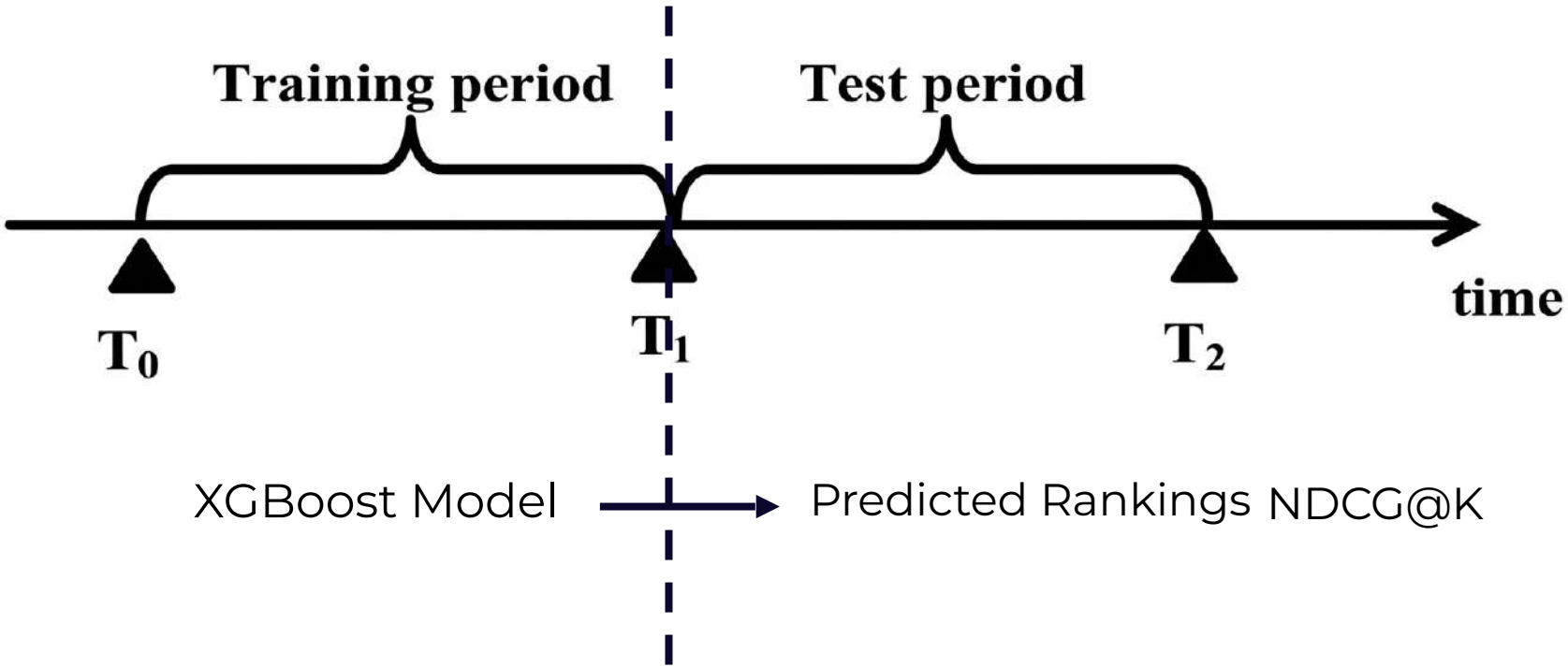

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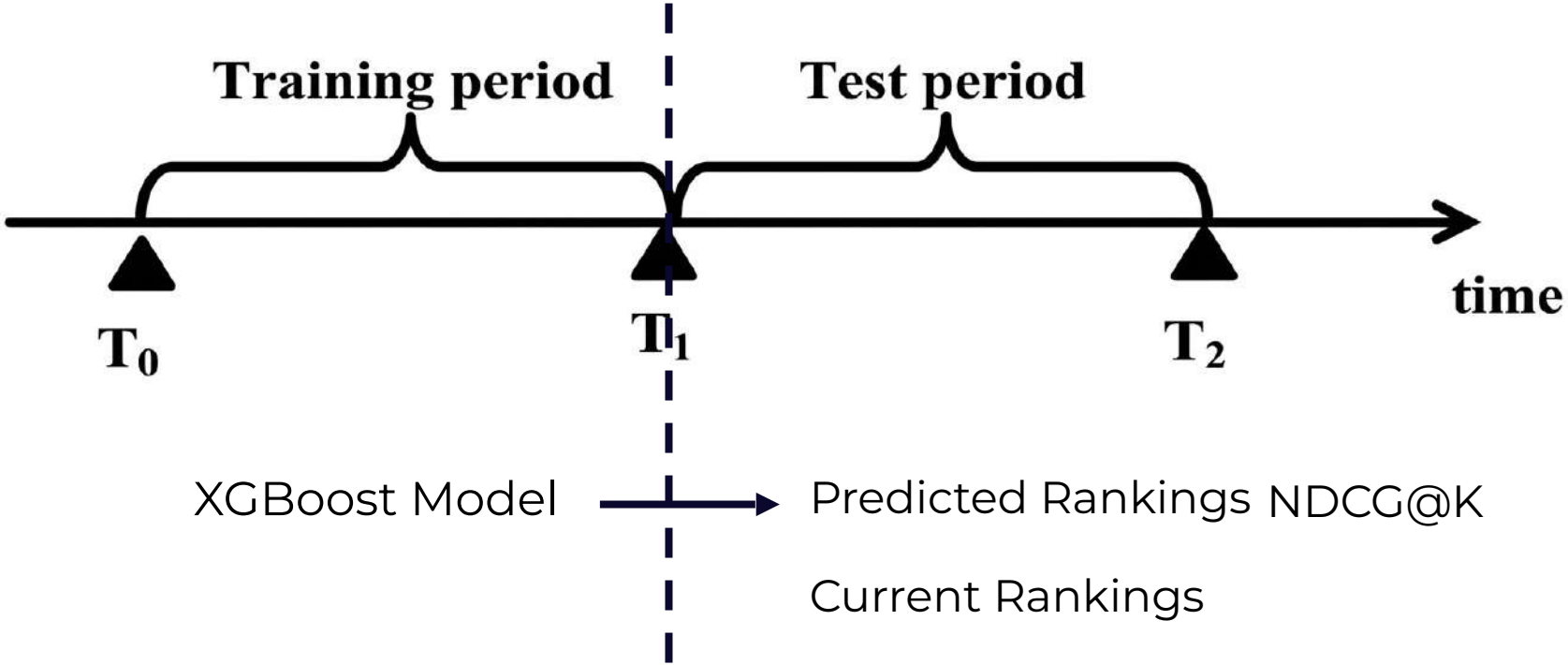

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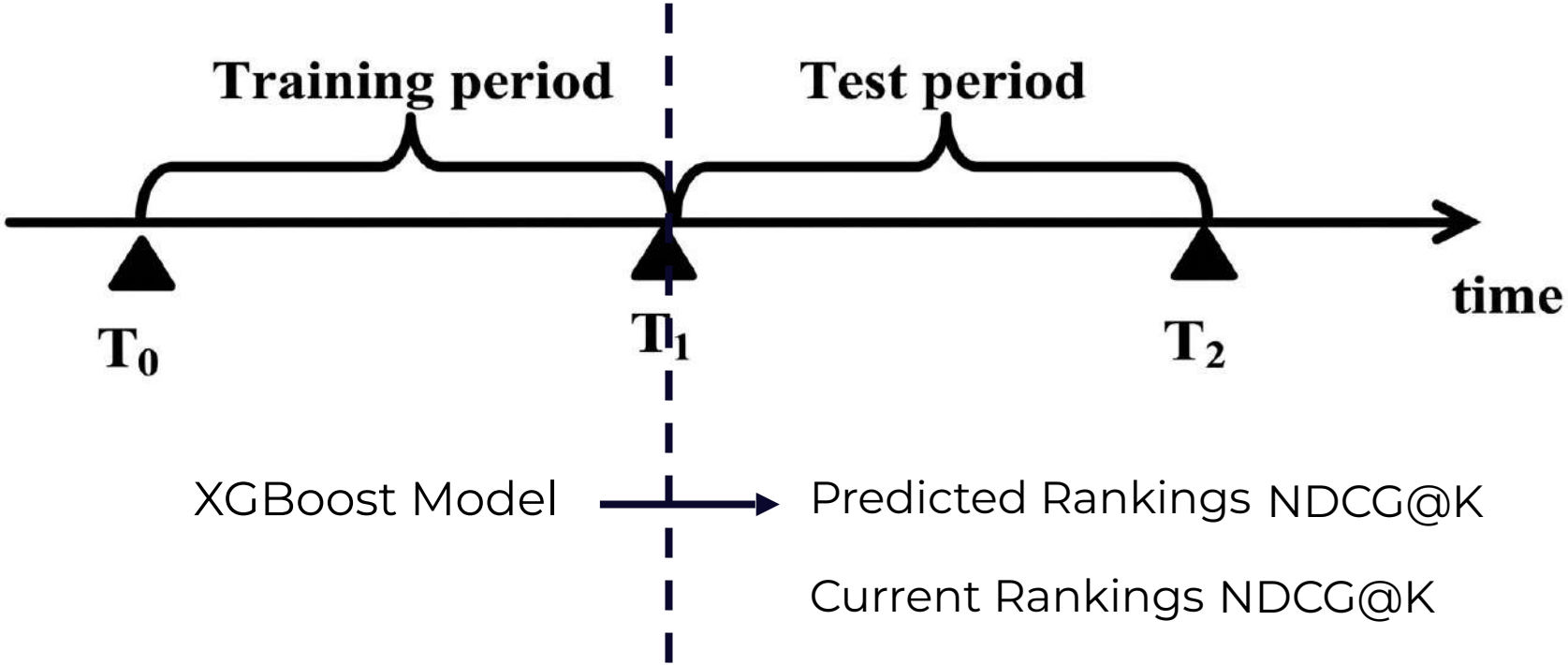

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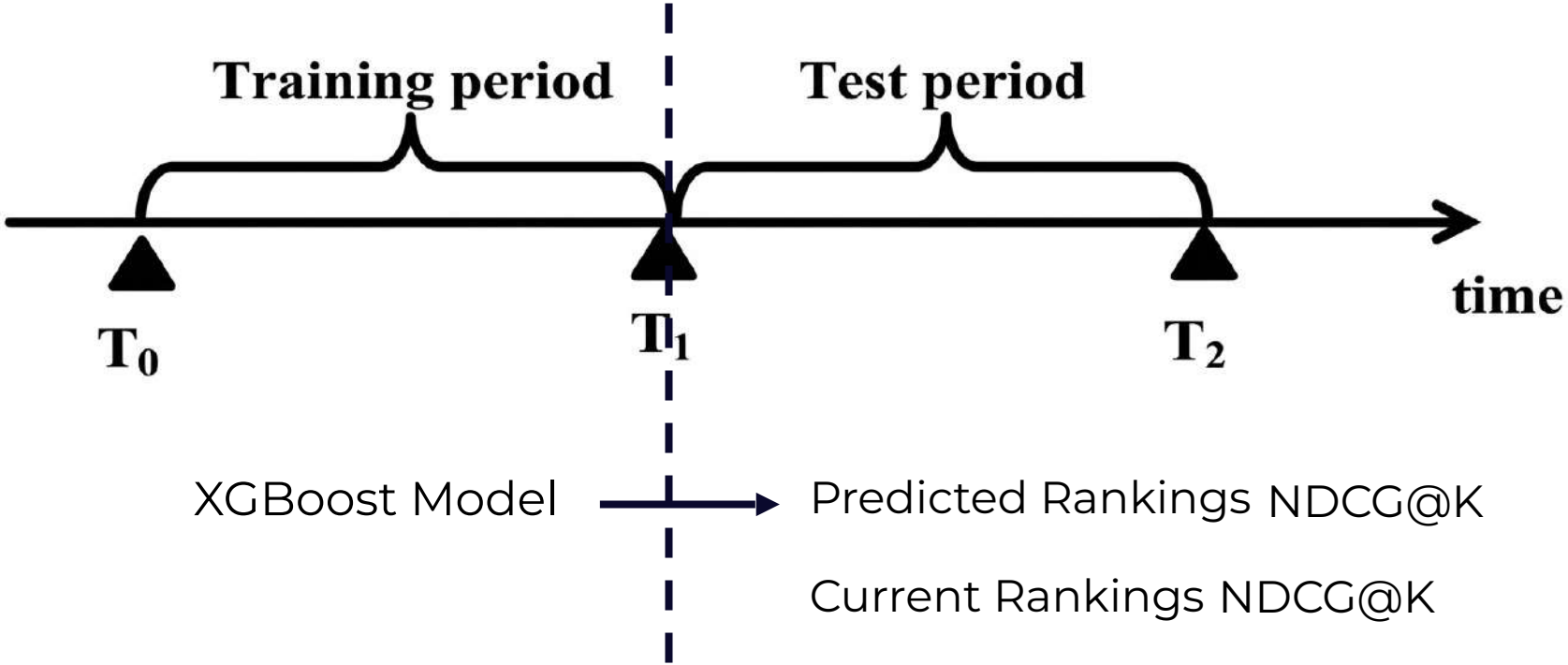
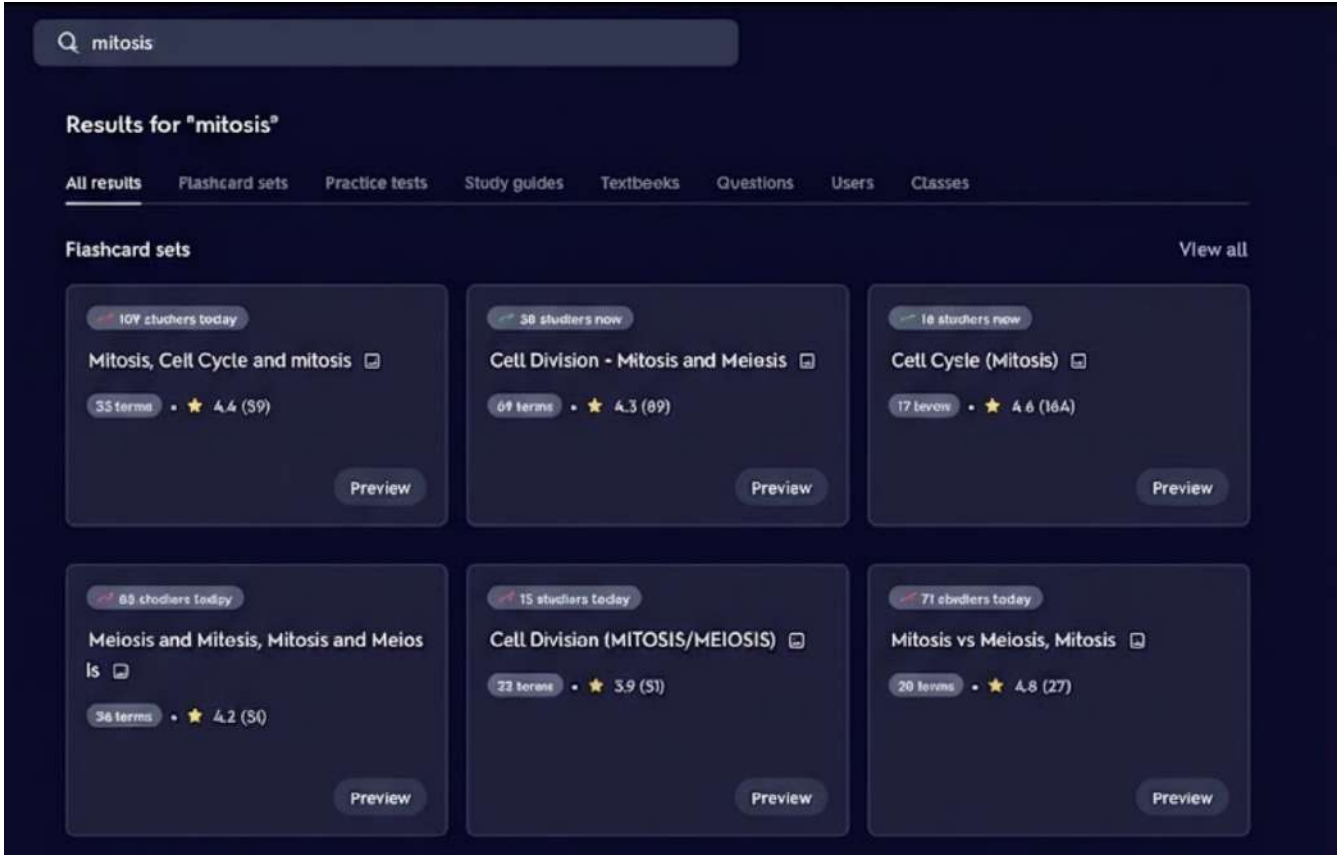


Lift?

Online Experimentation

Ranking Flashcard Sets



Control:
No Re-Ranking

Variant 1:
Re-rank N

Variant 2:
Re-rank 2*N

1. Elastic Documentation:
https://www.elastic.co/docs/solutions/search/ranking/learning-to-rank-ltr

2. XGBoost Documentation: https://xgboost.readthedocs.io/en/latest/tutorials/learning_to_rank.html

Thanks!

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Questions?

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